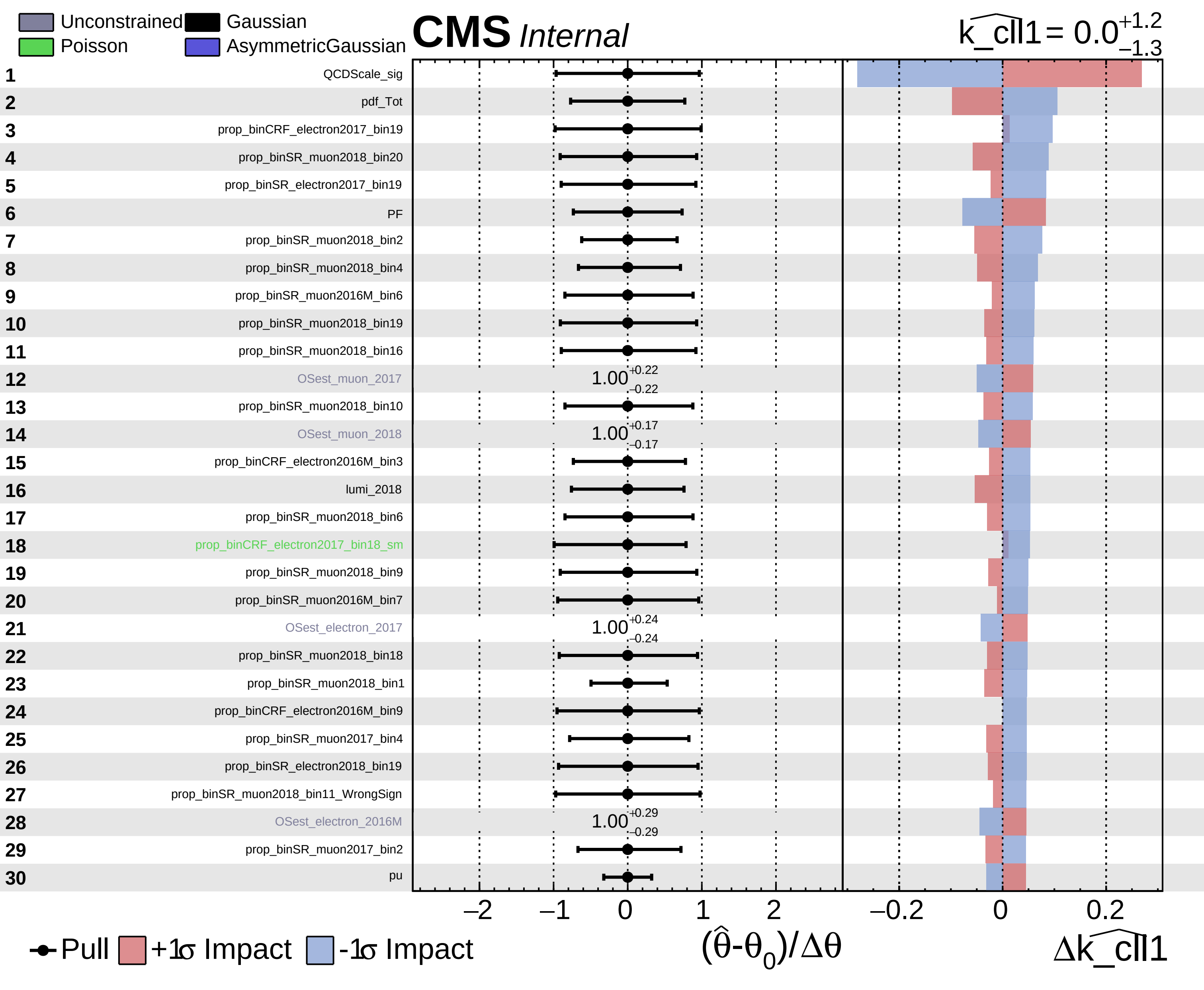
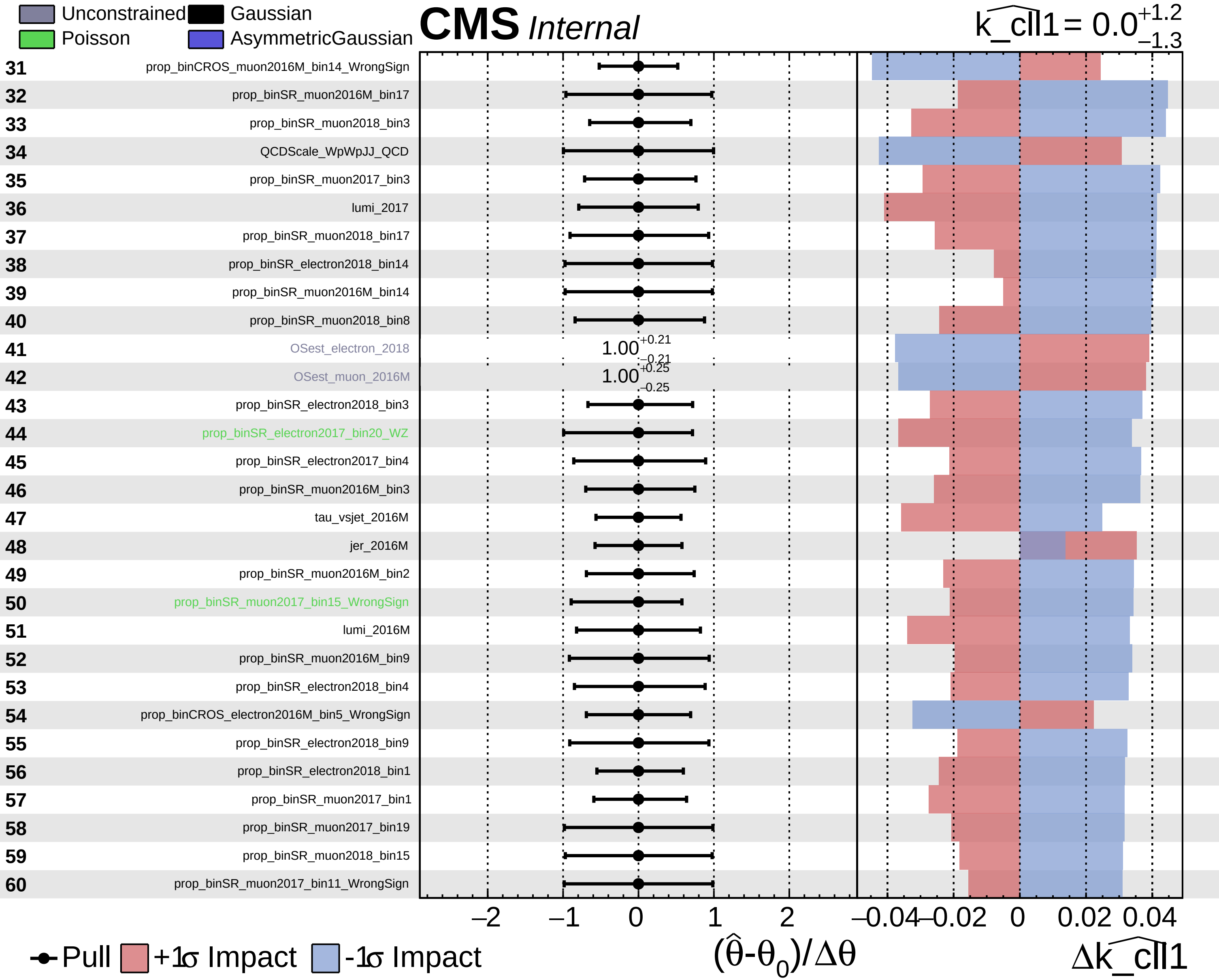
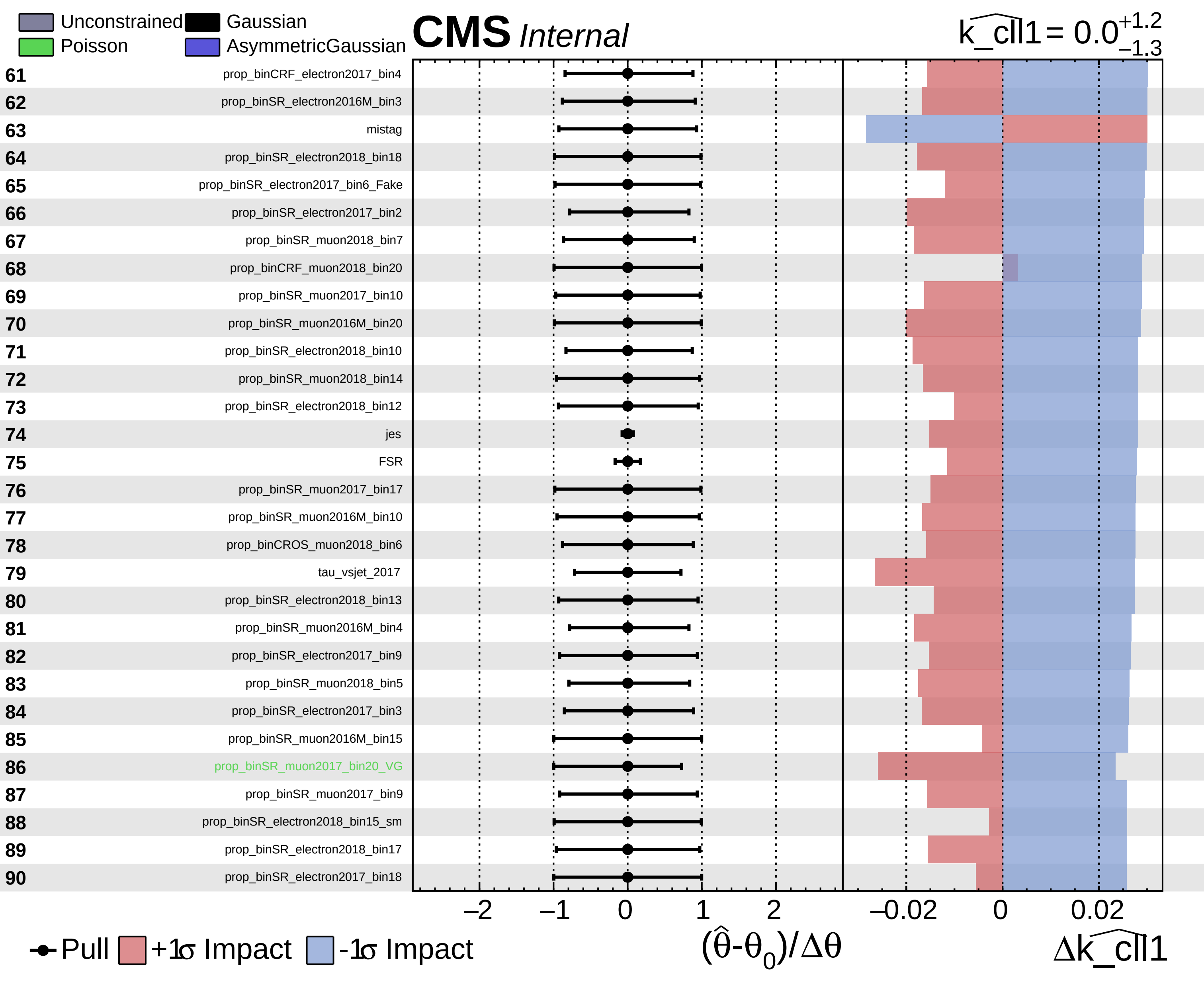


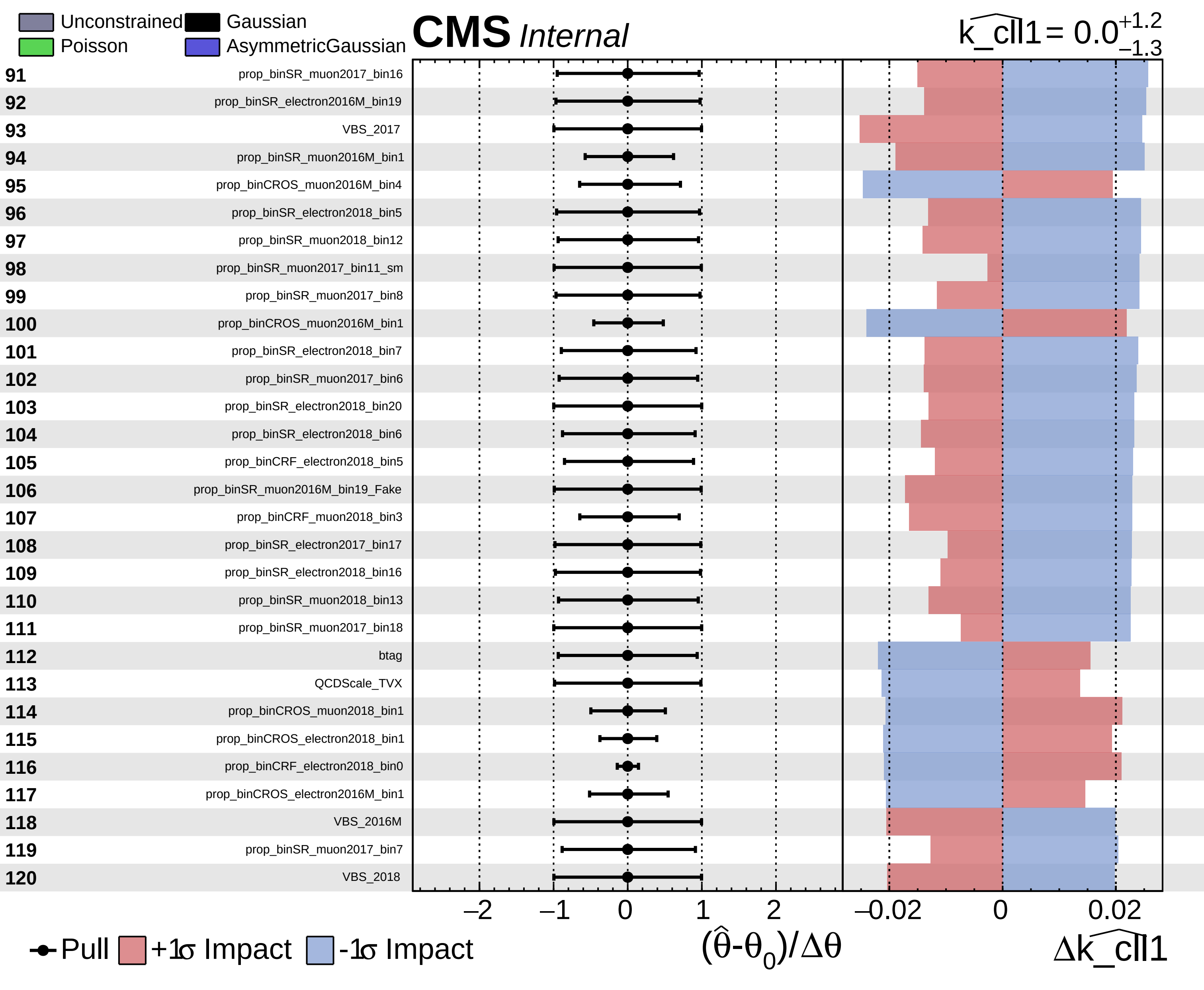
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
1 QCDScale_sig
2 pdf_Tot
3 prop_binCRF_electron2017_bin19
4 prop_binSR_muon2018_bin20
5 prop_binSR_electron2017_bin19
6 PF
7 prop_binSR_muon2018_bin2
8 prop_binSR_muon2018_bin4
9 prop_binSR_muon2016M_bin6
10 prop_binSR_muon2018_bin19
11 prop_binSR_muon2018_bin16
12 OSest_muon_2017 1.00 +0.22 −0.22
13 prop_binSR_muon2018_bin10
14 OSest_muon_2018 1.00 +0.17 −0.17
15 prop_binCRF_electron2016M_bin3
16 lumi_2018
17 prop_binSR_muon2018_bin6
18 prop_binCRF_electron2017_bin18_sm
19 prop_binSR_muon2018_bin9
20 prop_binSR_muon2016M_bin7
21 OSest_electron_2017 1.00 +0.24 −0.24
22 prop_binSR_muon2018_bin18
23 prop_binSR_muon2018_bin1
24 prop_binCRF_electron2016M_bin9
25 prop_binSR_muon2017_bin4
26 prop_binSR_electron2018_bin19
27 prop_binSR_muon2018_bin11_WrongSign
28 OSest_electron_2016M 1.00 +0.29 −0.29
29 prop_binSR_muon2017_bin2
30 pu
−2 −1 0 1 2
−0.2 0 0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cll1



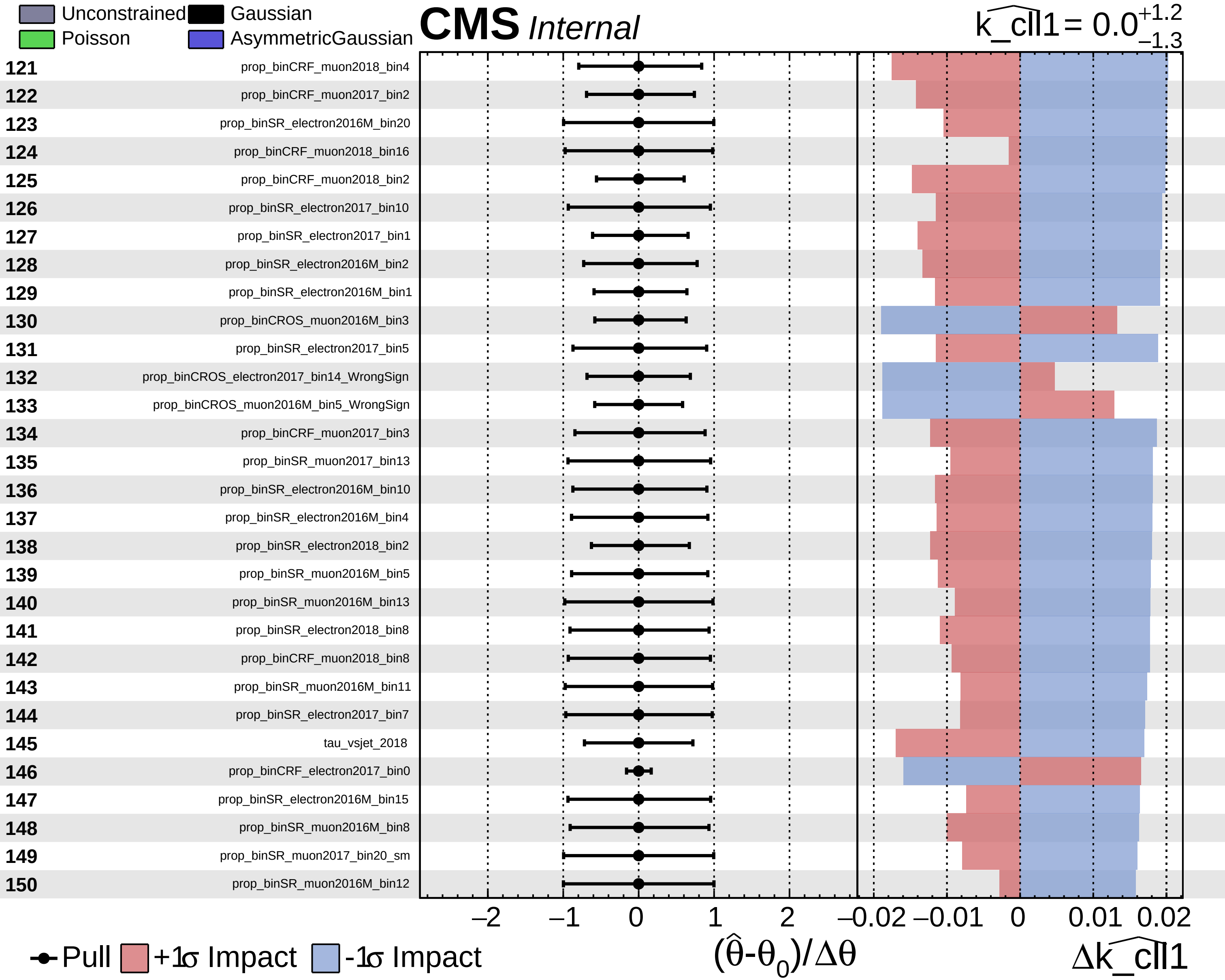
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
31 prop_binCROS_muon2016M_bin14_WrongSign
32 prop_binSR_muon2016M_bin17
33 prop_binSR_muon2018_bin3
34 QCDScale_WpWpJJ_QCD
35 prop_binSR_muon2017_bin3
36 lumi_2017
37 prop_binSR_muon2018_bin17
38 prop_binSR_electron2018_bin14
39 prop_binSR_muon2016M_bin14
40 prop_binSR_muon2018_bin8
41 OSest_electron_2018
1.00 +0.21 −0.21
42 OSest_muon_2016M
1.00 +0.25 −0.25
43 prop_binSR_electron2018_bin3
44 prop_binSR_electron2017_bin20_WZ
45 prop_binSR_electron2017_bin4
46 prop_binSR_muon2016M_bin3
47 tau_vsjet_2016M
48 jer_2016M
49 prop_binSR_muon2016M_bin2
50 prop_binSR_muon2017_bin15_WrongSign
51 lumi_2016M
52 prop_binSR_muon2016M_bin9
53 prop_binSR_electron2018_bin4
54 prop_binCROS_electron2016M_bin5_WrongSign
55 prop_binSR_electron2018_bin9
56 prop_binSR_electron2018_bin1
57 prop_binSR_muon2017_bin1
58 prop_binSR_muon2017_bin19
59 prop_binSR_muon2018_bin15
60 prop_binSR_muon2017_bin11_WrongSign
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cll1



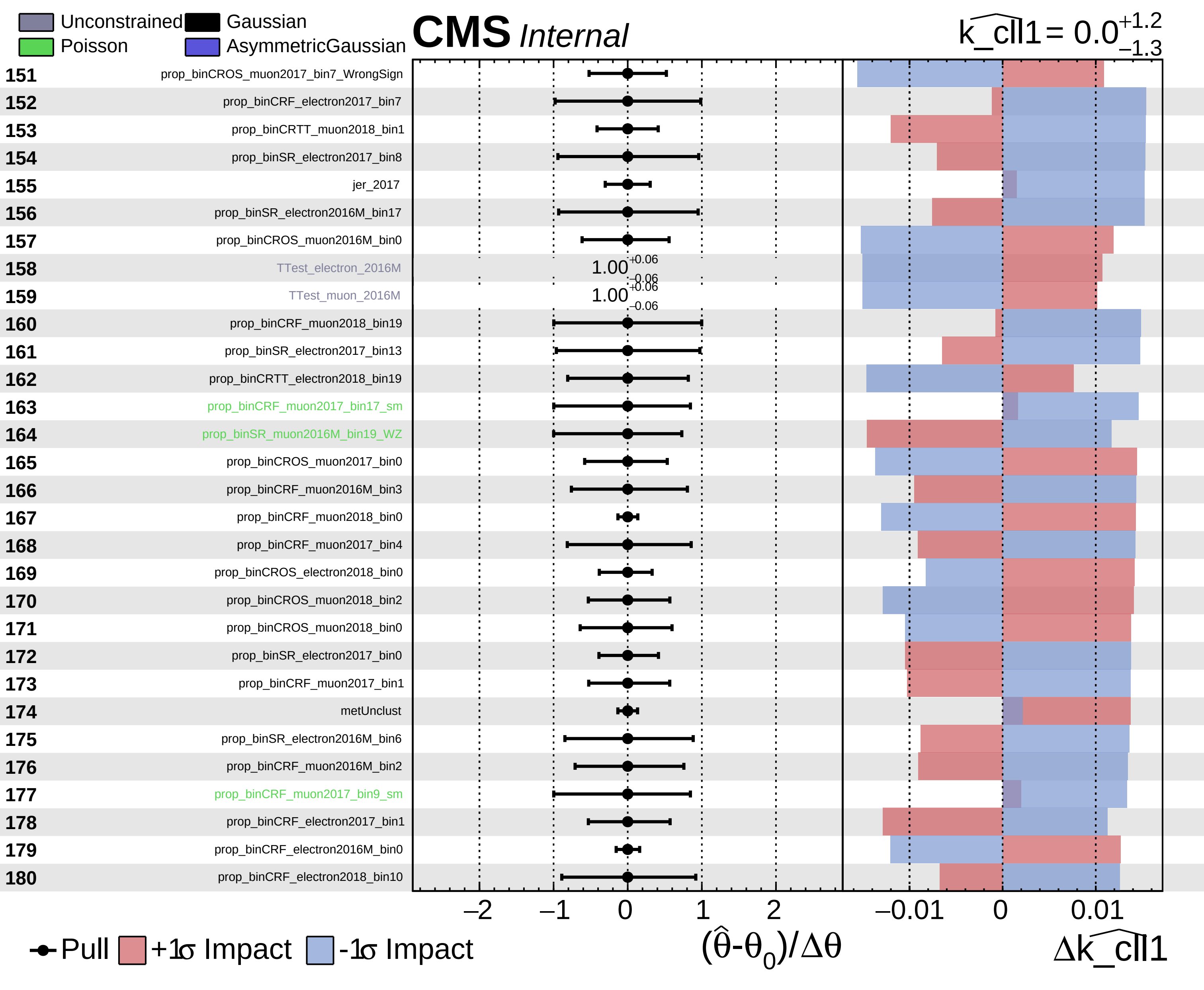
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cll1} = 0.0^{+1.2}_{-1.3}$
61 prop_binCRF_electron2017_bin4
62 prop_binSR_electron2016M_bin3
63 mistag
64 prop_binSR_electron2018_bin18
65 prop_binSR_electron2017_bin6_Fake
66 prop_binSR_electron2017_bin2
67 prop_binSR_muon2018_bin7
68 prop_binCRF_muon2018_bin20
69 prop_binSR_muon2017_bin10
70 prop_binSR_muon2016M_bin20
71 prop_binSR_electron2018_bin10
72 prop_binSR_muon2018_bin14
73 prop_binSR_electron2018_bin12
74 jes
75 FSR
76 prop_binSR_muon2017_bin17
77 prop_binSR_muon2016M_bin10
78 prop_binCROS_muon2018_bin6
79 tau_vsjet_2017
80 prop_binSR_electron2018_bin13
81 prop_binSR_muon2016M_bin4
82 prop_binSR_electron2017_bin9
83 prop_binSR_muon2018_bin5
84 prop_binSR_electron2017_bin3
85 prop_binSR_muon2016M_bin15
86 prop_binSR_muon2017_bin20_VG
87 prop_binSR_muon2017_bin9
88 prop_binSR_electron2018_bin15_sm
89 prop_binSR_electron2018_bin17
90 prop_binSR_electron2017_bin18
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cll1}$



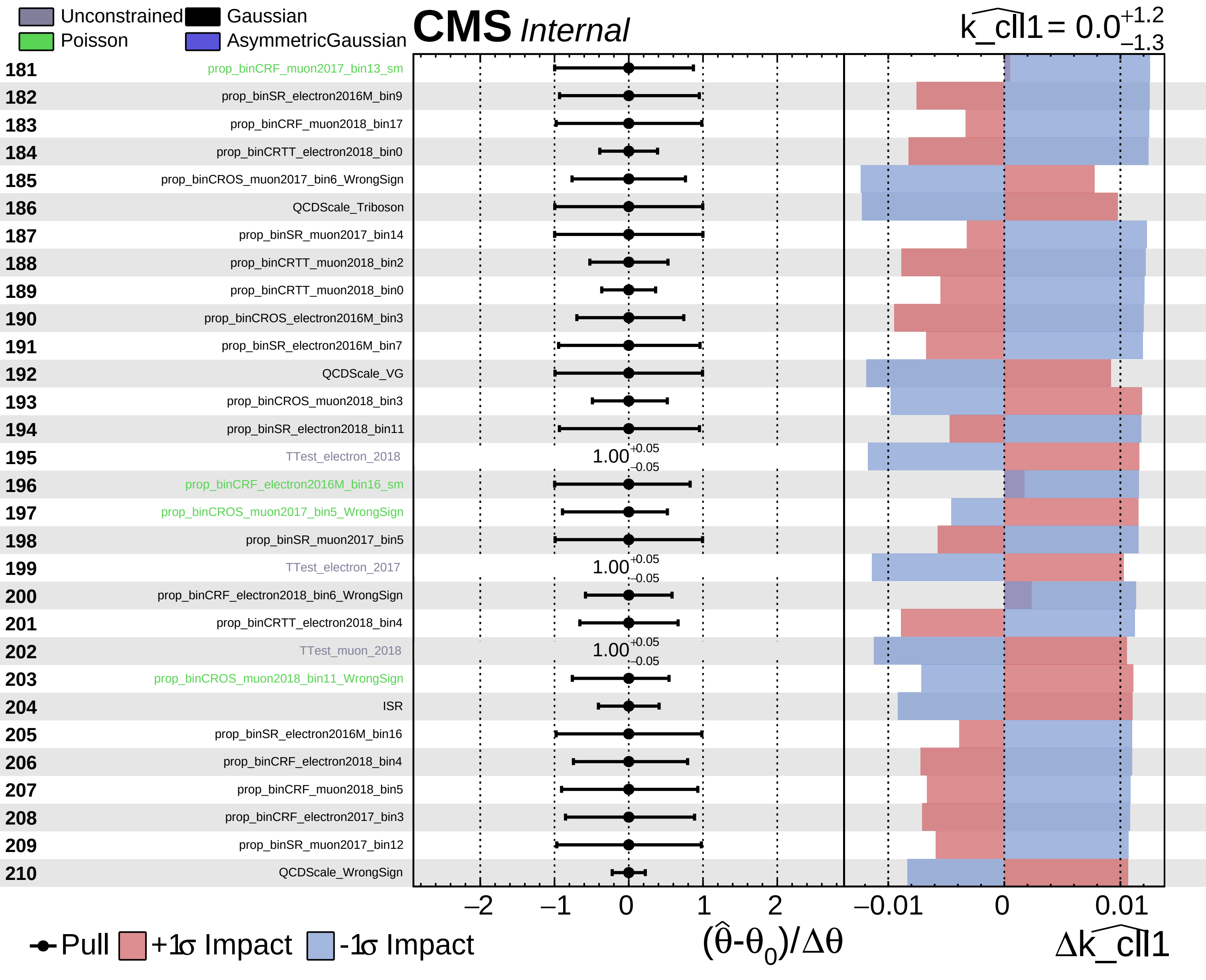
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cll1} = 0.0^{+1.2}_{-1.3}$
91 prop_binSR_muon2017_bin16
92 prop_binSR_electron2016M_bin19
93 VBS_2017
94 prop_binSR_muon2016M_bin1
95 prop_binCROS_muon2016M_bin4
96 prop_binSR_electron2018_bin5
97 prop_binSR_muon2018_bin12
98 prop_binSR_muon2017_bin11_sm
99 prop_binSR_muon2017_bin8
100 prop_binCROS_muon2016M_bin1
101 prop_binSR_electron2018_bin7
102 prop_binSR_muon2017_bin6
103 prop_binSR_electron2018_bin20
104 prop_binSR_electron2018_bin6
105 prop_binCRF_electron2018_bin5
106 prop_binSR_muon2016M_bin19_Fake
107 prop_binCRF_muon2018_bin3
108 prop_binSR_electron2017_bin17
109 prop_binSR_electron2018_bin16
110 prop_binSR_muon2018_bin13
111 prop_binSR_muon2017_bin18
112 btag
113 QCDScale_TVX
114 prop_binCROS_muon2018_bin1
115 prop_binCROS_electron2018_bin1
116 prop_binCRF_electron2018_bin0
117 prop_binCROS_electron2016M_bin1
118 VBS_2016M
119 prop_binSR_muon2017_bin7
120 VBS_2018
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cll1}$



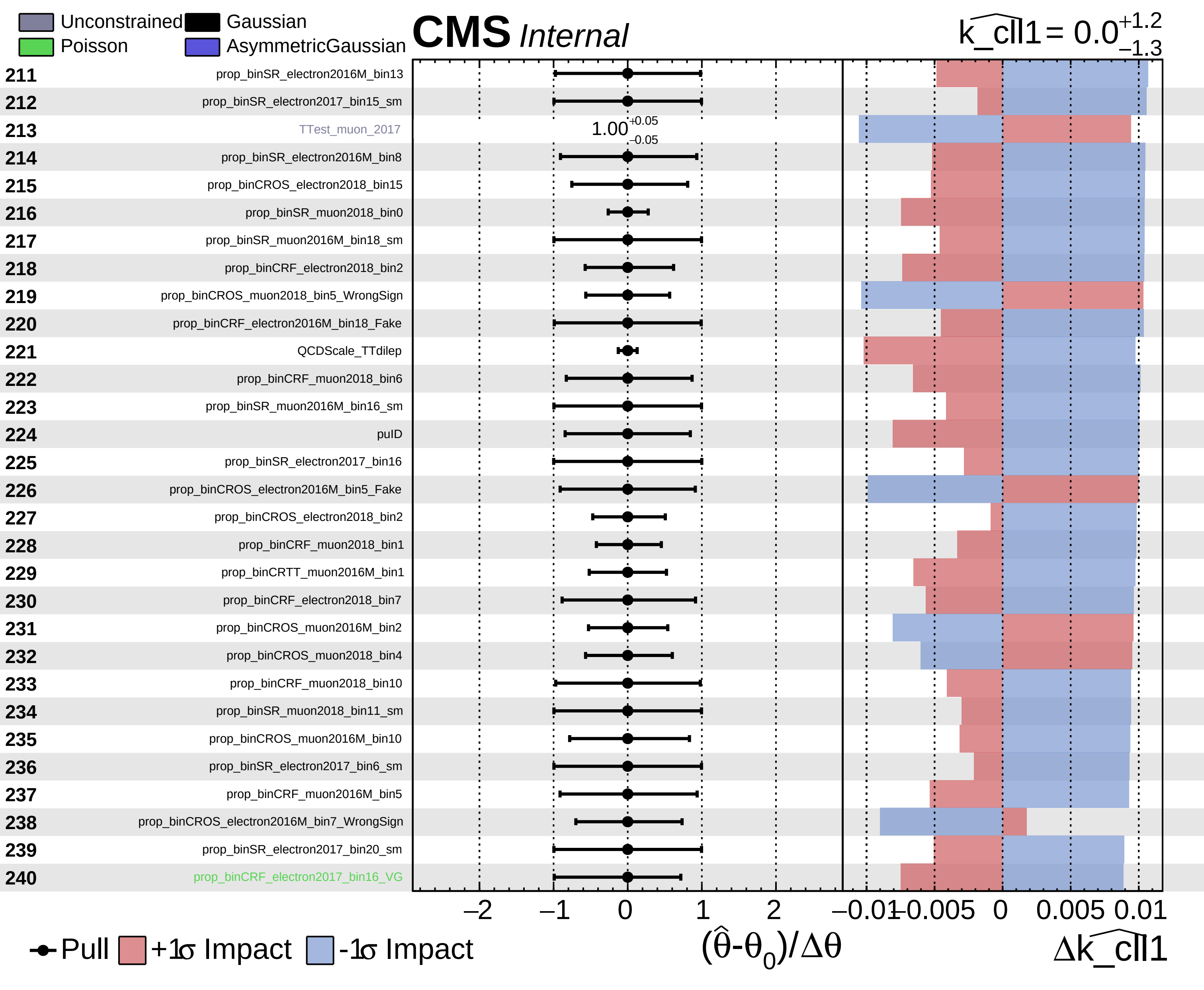
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cll1} = 0.0^{+1.2}_{-1.3}$
121 prop_binCRF_muon2018_bin4
122 prop_binCRF_muon2017_bin2
123 prop_binSR_electron2016M_bin20
124 prop_binCRF_muon2018_bin16
125 prop_binCRF_muon2018_bin2
126 prop_binSR_electron2017_bin10
127 prop_binSR_electron2017_bin1
128 prop_binSR_electron2016M_bin2
129 prop_binSR_electron2016M_bin1
130 prop_binCROS_muon2016M_bin3
131 prop_binSR_electron2017_bin5
132 prop_binCROS_electron2017_bin14_WrongSign
133 prop_binCROS_muon2016M_bin5_WrongSign
134 prop_binCRF_muon2017_bin3
135 prop_binSR_muon2017_bin13
136 prop_binSR_electron2016M_bin10
137 prop_binSR_electron2016M_bin4
138 prop_binSR_electron2018_bin2
139 prop_binSR_muon2016M_bin5
140 prop_binSR_muon2016M_bin13
141 prop_binSR_electron2018_bin8
142 prop_binCRF_muon2018_bin8
143 prop_binSR_muon2016M_bin11
144 prop_binSR_electron2017_bin7
145 tau_vsjet_2018
146 prop_binCRF_electron2017_bin0
147 prop_binSR_electron2016M_bin15
148 prop_binSR_muon2016M_bin8
149 prop_binSR_muon2017_bin20_sm
150 prop_binSR_muon2016M_bin12
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cll1}$



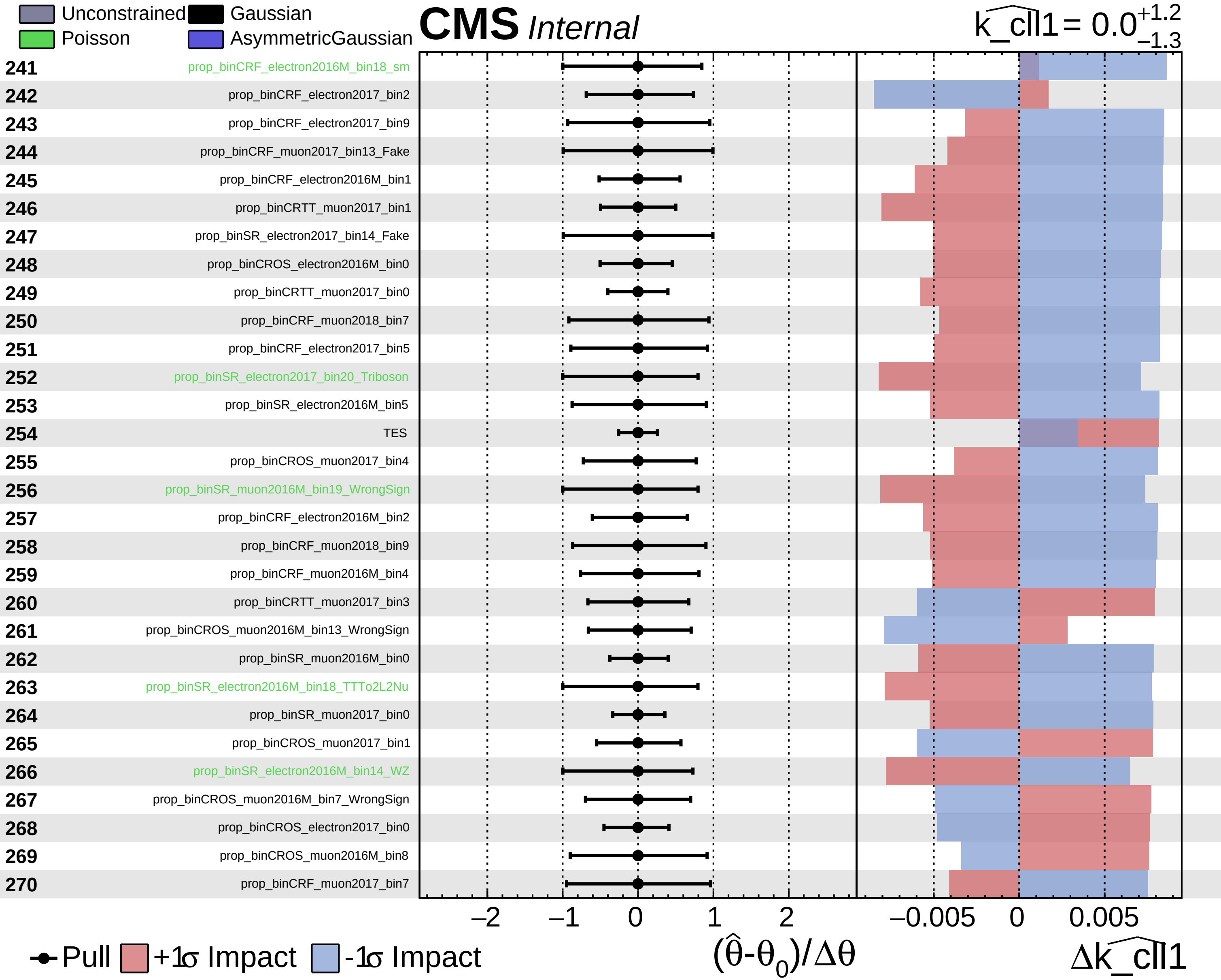
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
151 prop_binCROS_muon2017_bin7_WrongSign
152 prop_binCRF_electron2017_bin7
153 prop_binCRTT_muon2018_bin1
154 prop_binSR_electron2017_bin8
155 jer_2017
156 prop_binSR_electron2016M_bin17
157 prop_binCROS_muon2016M_bin0
158 TTest_electron_2016M 1.00 +0.06 −0.06
159 TTest_muon_2016M 1.00 +0.06 −0.06
160 prop_binCRF_muon2018_bin19
161 prop_binSR_electron2017_bin13
162 prop_binCRTT_electron2018_bin19
163 prop_binCRF_muon2017_bin17_sm
164 prop_binSR_muon2016M_bin19_WZ
165 prop_binCROS_muon2017_bin0
166 prop_binCRF_muon2016M_bin3
167 prop_binCRF_muon2018_bin0
168 prop_binCRF_muon2017_bin4
169 prop_binCROS_electron2018_bin0
170 prop_binCROS_muon2018_bin2
171 prop_binCROS_muon2018_bin0
172 prop_binSR_electron2017_bin0
173 prop_binCRF_muon2017_bin1
174 metUnclust
175 prop_binSR_electron2016M_bin6
176 prop_binCRF_muon2016M_bin2
177 prop_binCRF_muon2017_bin9_sm
178 prop_binCRF_electron2017_bin1
179 prop_binCRF_electron2016M_bin0
180 prop_binCRF_electron2018_bin10
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cll1



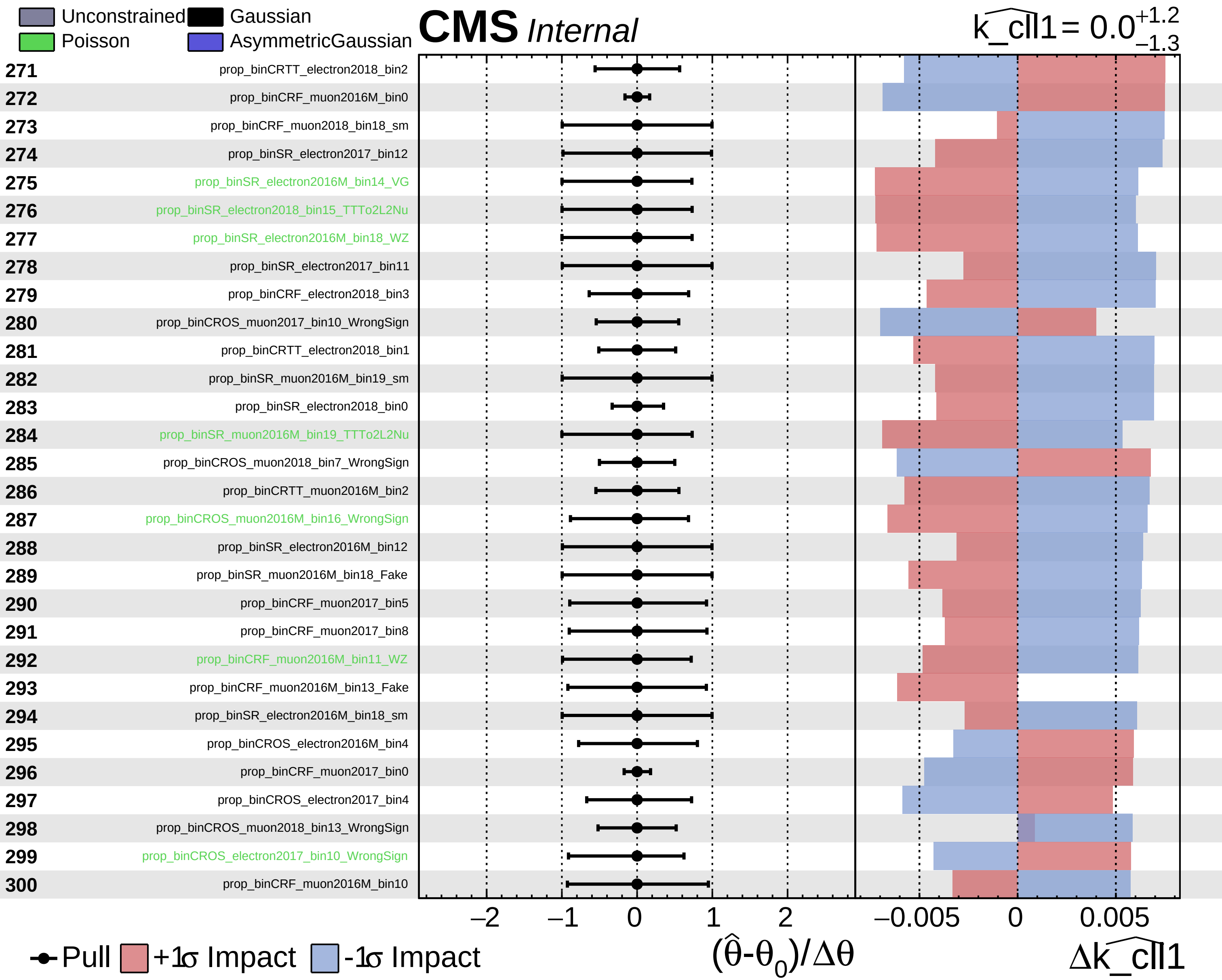
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
181 prop_binCRF_muon2017_bin13_sm
182 prop_binSR_electron2016M_bin9
183 prop_binCRF_muon2018_bin17
184 prop_binCRTT_electron2018_bin0
185 prop_binCROS_muon2017_bin6_WrongSign
186 QCDScale_Triboson
187 prop_binSR_muon2017_bin14
188 prop_binCRTT_muon2018_bin2
189 prop_binCRTT_muon2018_bin0
190 prop_binCROS_electron2016M_bin3
191 prop_binSR_electron2016M_bin7
192 QCDScale_VG
193 prop_binCROS_muon2018_bin3
194 prop_binSR_electron2018_bin11
195 TTest_electron_2018 1.00 +0.05 −0.05
196 prop_binCRF_electron2016M_bin16_sm
197 prop_binCROS_muon2017_bin5_WrongSign
198 prop_binSR_muon2017_bin5
199 TTest_electron_2017 1.00 +0.05 −0.05
200 prop_binCRF_electron2018_bin6_WrongSign
201 prop_binCRTT_electron2018_bin4
202 TTest_muon_2018 1.00 +0.05 −0.05
203 prop_binCROS_muon2018_bin11_WrongSign
204 ISR
205 prop_binSR_electron2016M_bin16
206 prop_binCRF_electron2018_bin4
207 prop_binCRF_muon2018_bin5
208 prop_binCRF_electron2017_bin3
209 prop_binSR_muon2017_bin12
210 QCDScale_WrongSign
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cll1



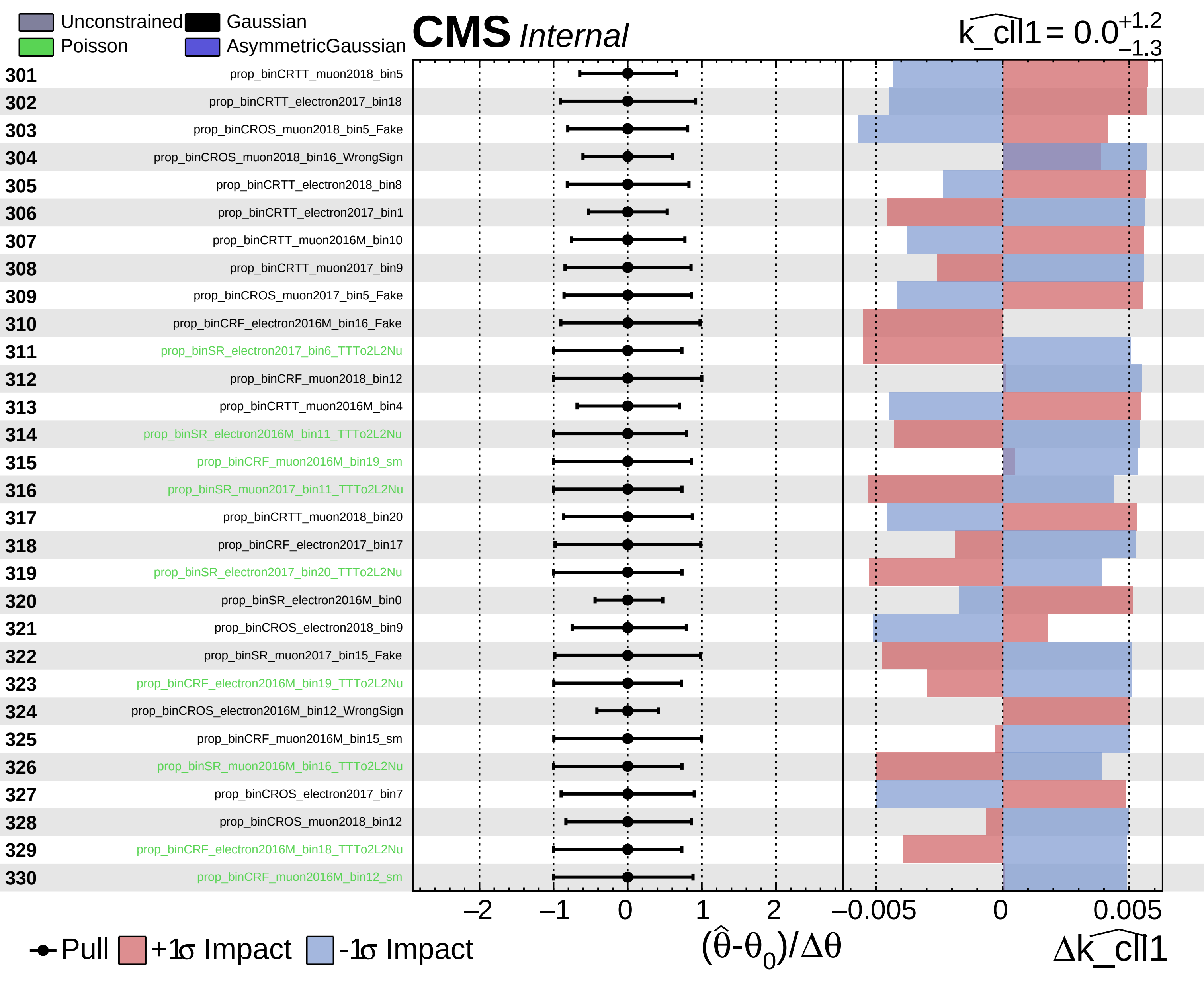
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
211 prop_binSR_electron2016M_bin13
212 prop_binSR_electron2017_bin15_sm
213 TTest_muon_2017
1.00 +0.05 −0.05
214 prop_binSR_electron2016M_bin8
215 prop_binCROS_electron2018_bin15
216 prop_binSR_muon2018_bin0
217 prop_binSR_muon2016M_bin18_sm
218 prop_binCRF_electron2018_bin2
219 prop_binCROS_muon2018_bin5_WrongSign
220 prop_binCRF_electron2016M_bin18_Fake
221 QCDScale_TTdilep
222 prop_binCRF_muon2018_bin6
223 prop_binSR_muon2016M_bin16_sm
224 puID
225 prop_binSR_electron2017_bin16
226 prop_binCROS_electron2016M_bin5_Fake
227 prop_binCROS_electron2018_bin2
228 prop_binCRF_muon2018_bin1
229 prop_binCRTT_muon2016M_bin1
230 prop_binCRF_electron2018_bin7
231 prop_binCROS_muon2016M_bin2
232 prop_binCROS_muon2018_bin4
233 prop_binCRF_muon2018_bin10
234 prop_binSR_muon2018_bin11_sm
235 prop_binCROS_muon2016M_bin10
236 prop_binSR_electron2017_bin6_sm
237 prop_binCRF_muon2018_bin5
238 prop_binCROS_electron2016M_bin7_WrongSign
239 prop_binSR_electron2017_bin20_sm
240 prop_binCRF_electron2017_bin16_VG
−2
−1
0
1
2
−0.01
−0.005
0
0.005
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cll1



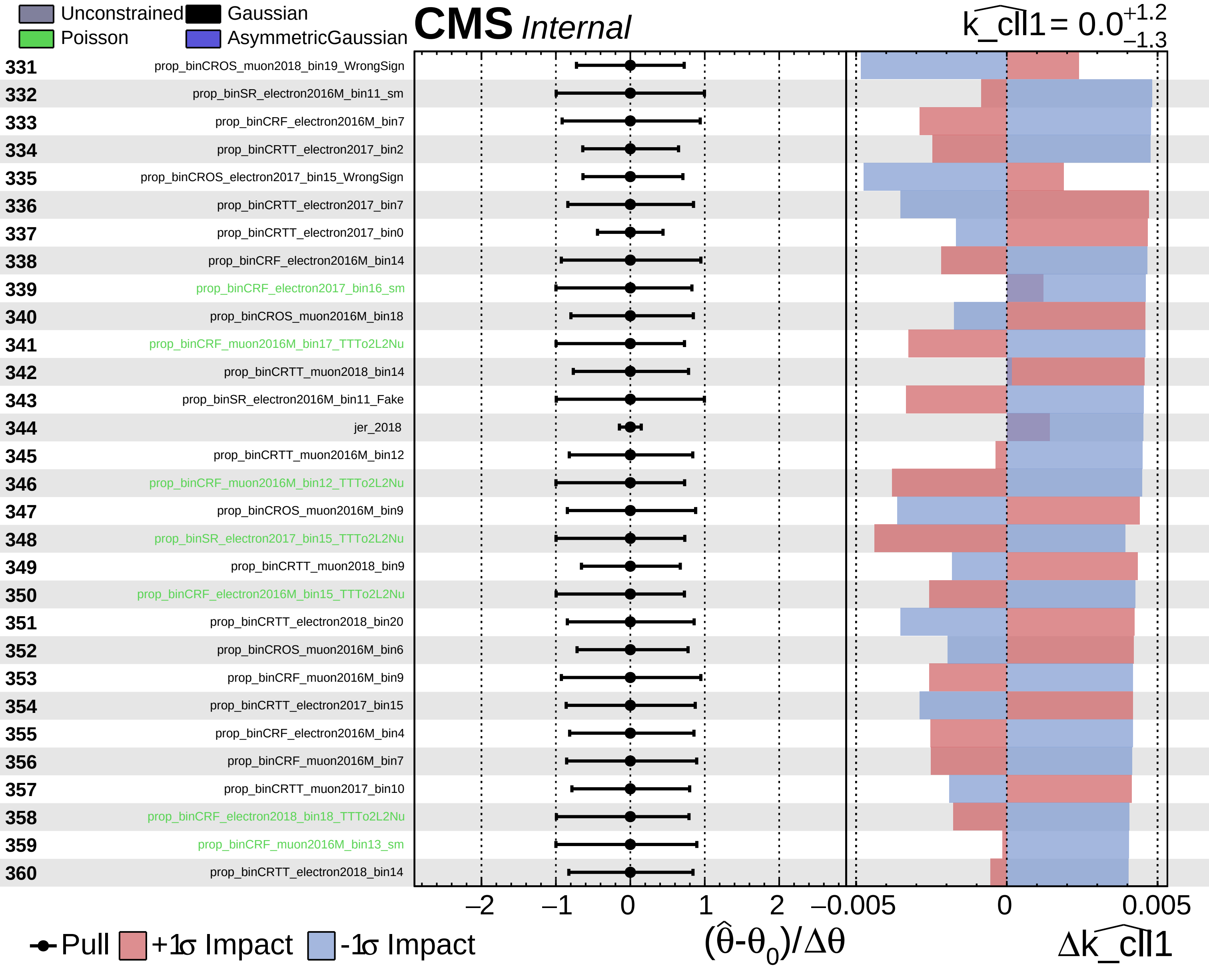
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
241 prop_binCRF_electron2016M_bin18_sm
242 prop_binCRF_electron2017_bin2
243 prop_binCRF_electron2017_bin9
244 prop_binCRF_muon2017_bin13_Fake
245 prop_binCRF_electron2016M_bin1
246 prop_binCRTT_muon2017_bin1
247 prop_binSR_electron2017_bin14_Fake
248 prop_binCROS_electron2016M_bin0
249 prop_binCRTT_muon2017_bin0
250 prop_binCRF_muon2018_bin7
251 prop_binCRF_electron2017_bin5
252 prop_binSR_electron2017_bin20_Triboson
253 prop_binSR_electron2016M_bin5
254 TES
255 prop_binCROS_muon2017_bin4
256 prop_binSR_muon2016M_bin19_WrongSign
257 prop_binCRF_electron2016M_bin2
258 prop_binCRF_muon2018_bin9
259 prop_binCRF_muon2016M_bin4
260 prop_binCRTT_muon2017_bin3
261 prop_binCROS_muon2016M_bin13_WrongSign
262 prop_binSR_muon2016M_bin0
263 prop_binSR_electron2016M_bin18_TTTo2L2Nu
264 prop_binSR_muon2017_bin0
265 prop_binCROS_muon2017_bin1
266 prop_binSR_electron2016M_bin14_WZ
267 prop_binCROS_muon2016M_bin7_WrongSign
268 prop_binCROS_electron2017_bin0
269 prop_binCROS_muon2016M_bin8
270 prop_binCRF_muon2017_bin7
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cll1



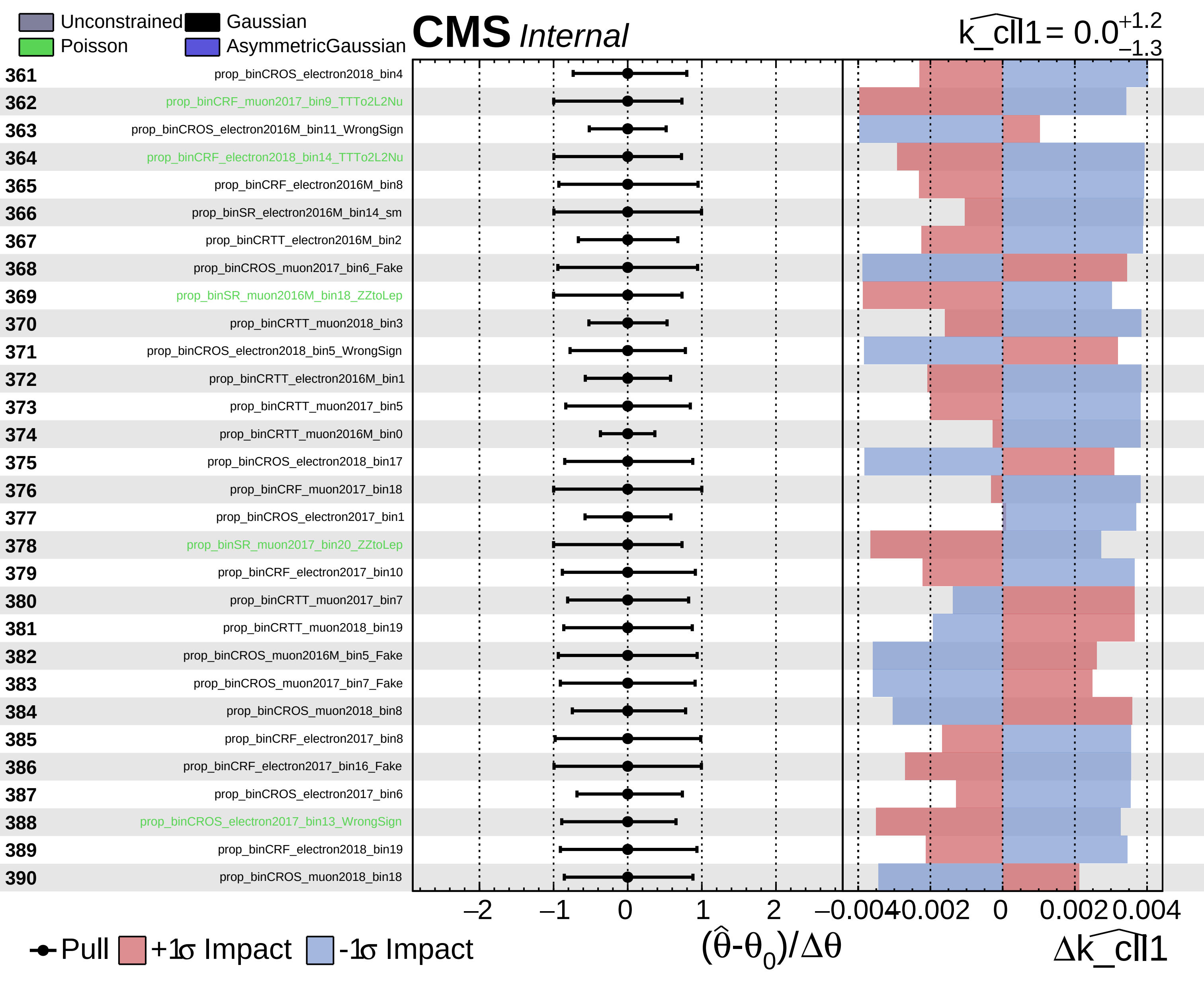
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
271 prop_binCRTT_electron2018_bin2
272 prop_binCRF_muon2016M_bin0
273 prop_binCRF_muon2018_bin18_sm
274 prop_binSR_electron2017_bin12
275 prop_binSR_electron2016M_bin14_VG
276 prop_binSR_electron2018_bin15_TTTo2L2Nu
277 prop_binSR_electron2016M_bin18_WZ
278 prop_binSR_electron2017_bin11
279 prop_binCRF_electron2018_bin3
280 prop_binCROS_muon2017_bin10_WrongSign
281 prop_binCRTT_electron2018_bin1
282 prop_binSR_muon2016M_bin19_sm
283 prop_binSR_electron2018_bin0
284 prop_binSR_muon2016M_bin19_TTTo2L2Nu
285 prop_binCROS_muon2018_bin7_WrongSign
286 prop_binCRTT_muon2016M_bin2
287 prop_binCROS_muon2016M_bin16_WrongSign
288 prop_binSR_electron2016M_bin12
289 prop_binSR_muon2016M_bin18_Fake
290 prop_binCRF_muon2017_bin5
291 prop_binCRF_muon2017_bin8
292 prop_binCRF_muon2016M_bin11_WZ
293 prop_binCRF_muon2016M_bin13_Fake
294 prop_binSR_electron2016M_bin18_sm
295 prop_binCROS_electron2016M_bin4
296 prop_binCRF_muon2017_bin0
297 prop_binCROS_electron2017_bin4
298 prop_binCROS_muon2018_bin13_WrongSign
299 prop_binCROS_electron2017_bin10_WrongSign
300 prop_binCRF_muon2016M_bin10
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cll1}$



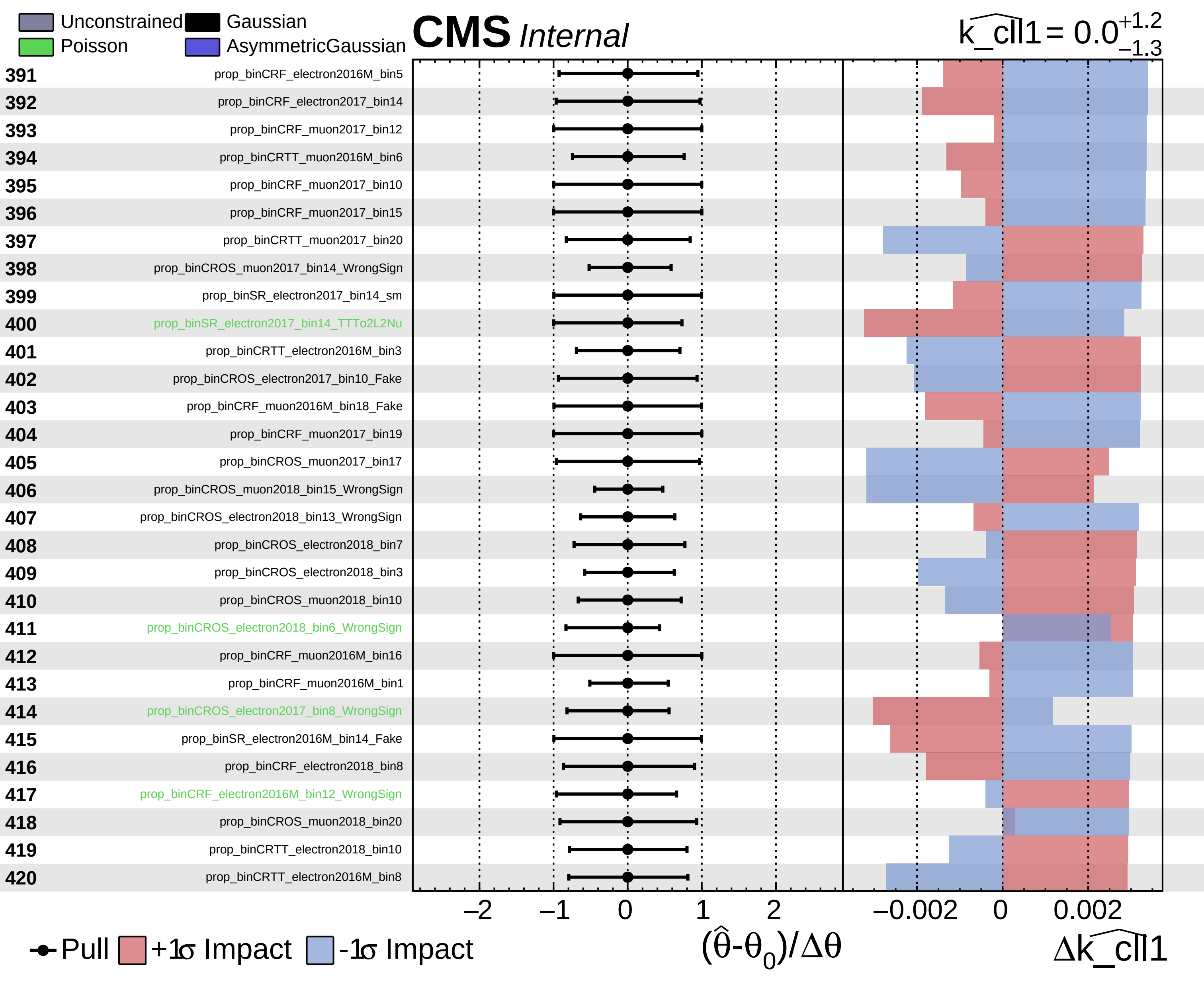
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
301 prop_binCRTT_muon2018_bin5
302 prop_binCRTT_electron2017_bin18
303 prop_binCROS_muon2018_bin5_Fake
304 prop_binCROS_muon2018_bin16_WrongSign
305 prop_binCRTT_electron2018_bin8
306 prop_binCRTT_electron2017_bin1
307 prop_binCRTT_muon2016M_bin10
308 prop_binCRTT_muon2017_bin9
309 prop_binCROS_muon2017_bin5_Fake
310 prop_binCRF_electron2016M_bin16_Fake
311 prop_binSR_electron2017_bin6_TTTo2L2Nu
312 prop_binCRF_muon2018_bin12
313 prop_binCRTT_muon2016M_bin4
314 prop_binSR_electron2016M_bin11_TTTo2L2Nu
315 prop_binCRF_muon2016M_bin19_sm
316 prop_binSR_muon2017_bin11_TTTo2L2Nu
317 prop_binCRTT_muon2018_bin20
318 prop_binCRF_electron2017_bin17
319 prop_binSR_electron2017_bin20_TTTo2L2Nu
320 prop_binSR_electron2016M_bin0
321 prop_binCROS_electron2018_bin9
322 prop_binSR_muon2017_bin15_Fake
323 prop_binCRF_electron2016M_bin19_TTTo2L2Nu
324 prop_binCROS_electron2016M_bin12_WrongSign
325 prop_binCRF_muon2016M_bin15_sm
326 prop_binSR_muon2016M_bin16_TTTo2L2Nu
327 prop_binCROS_electron2017_bin7
328 prop_binCROS_muon2018_bin12
329 prop_binCRF_electron2016M_bin18_TTTo2L2Nu
330 prop_binCRF_muon2016M_bin12_sm
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cll1

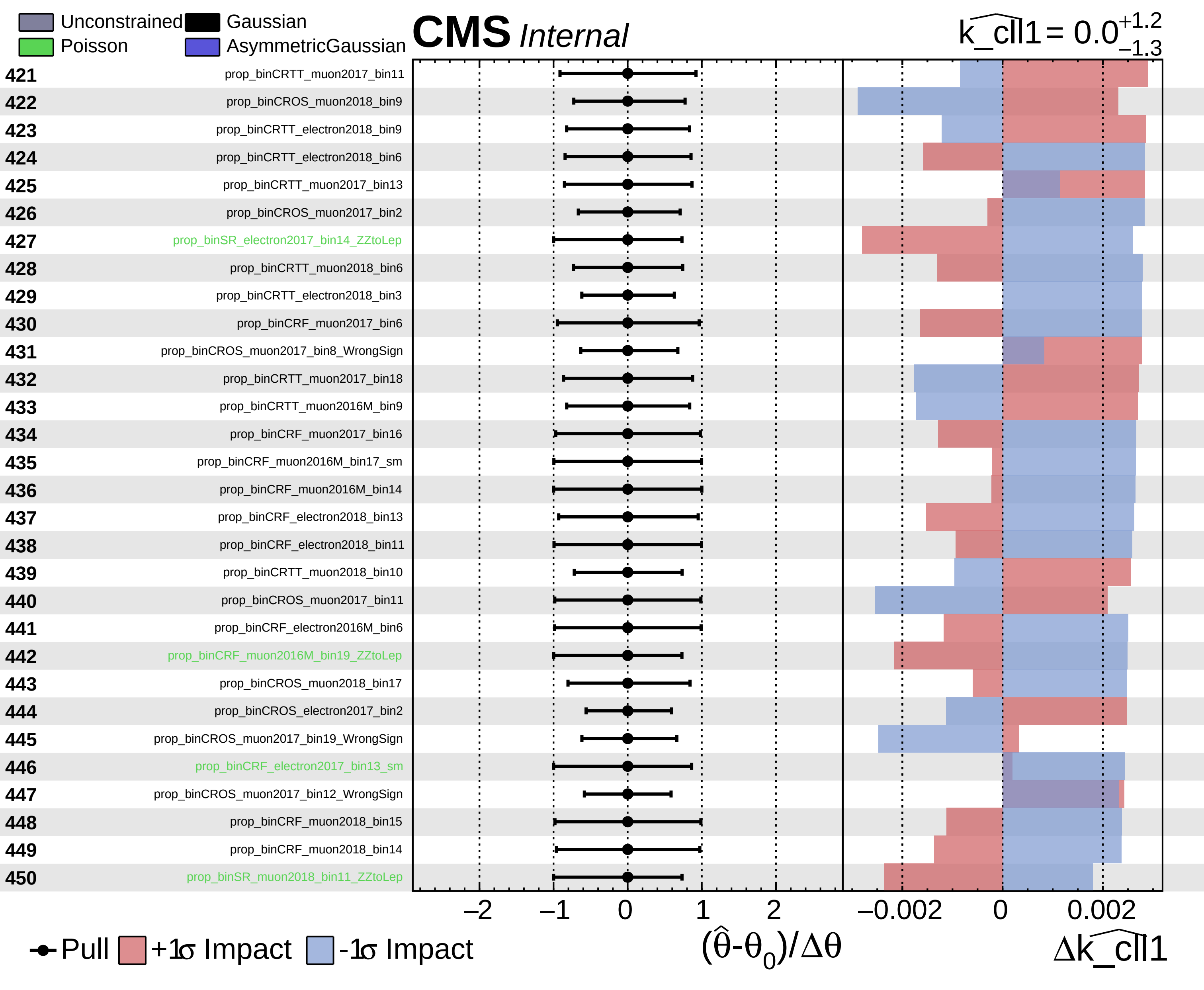


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cll1} = 0.0^{+1.2}_{-1.3}$
331 prop_binCROS_muon2018_bin19_WrongSign
332 prop_binSR_electron2016M_bin11_sm
333 prop_binCRF_electron2016M_bin7
334 prop_binCRTT_electron2017_bin2
335 prop_binCROS_electron2017_bin15_WrongSign
336 prop_binCRTT_electron2017_bin7
337 prop_binCRTT_electron2017_bin0
338 prop_binCRF_electron2016M_bin14
339 prop_binCRF_electron2017_bin16_sm
340 prop_binCROS_muon2016M_bin18
341 prop_binCRF_muon2016M_bin17_TTTo2L2Nu
342 prop_binCRTT_muon2018_bin14
343 prop_binSR_electron2016M_bin11_Fake
344 jer_2018
345 prop_binCRTT_muon2016M_bin12
346 prop_binCRF_muon2016M_bin12_TTTo2L2Nu
347 prop_binCROS_muon2016M_bin9
348 prop_binSR_electron2017_bin15_TTTo2L2Nu
349 prop_binCRTT_muon2018_bin9
350 prop_binCRF_electron2016M_bin15_TTTo2L2Nu
351 prop_binCRTT_electron2018_bin20
352 prop_binCROS_muon2016M_bin6
353 prop_binCRF_muon2016M_bin9
354 prop_binCRTT_electron2017_bin15
355 prop_binCRF_electron2016M_bin4
356 prop_binCRF_muon2016M_bin7
357 prop_binCRTT_muon2017_bin10
358 prop_binCRF_electron2018_bin18_TTTo2L2Nu
359 prop_binCRF_muon2016M_bin13_sm
360 prop_binCRTT_electron2018_bin14
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cll1}$

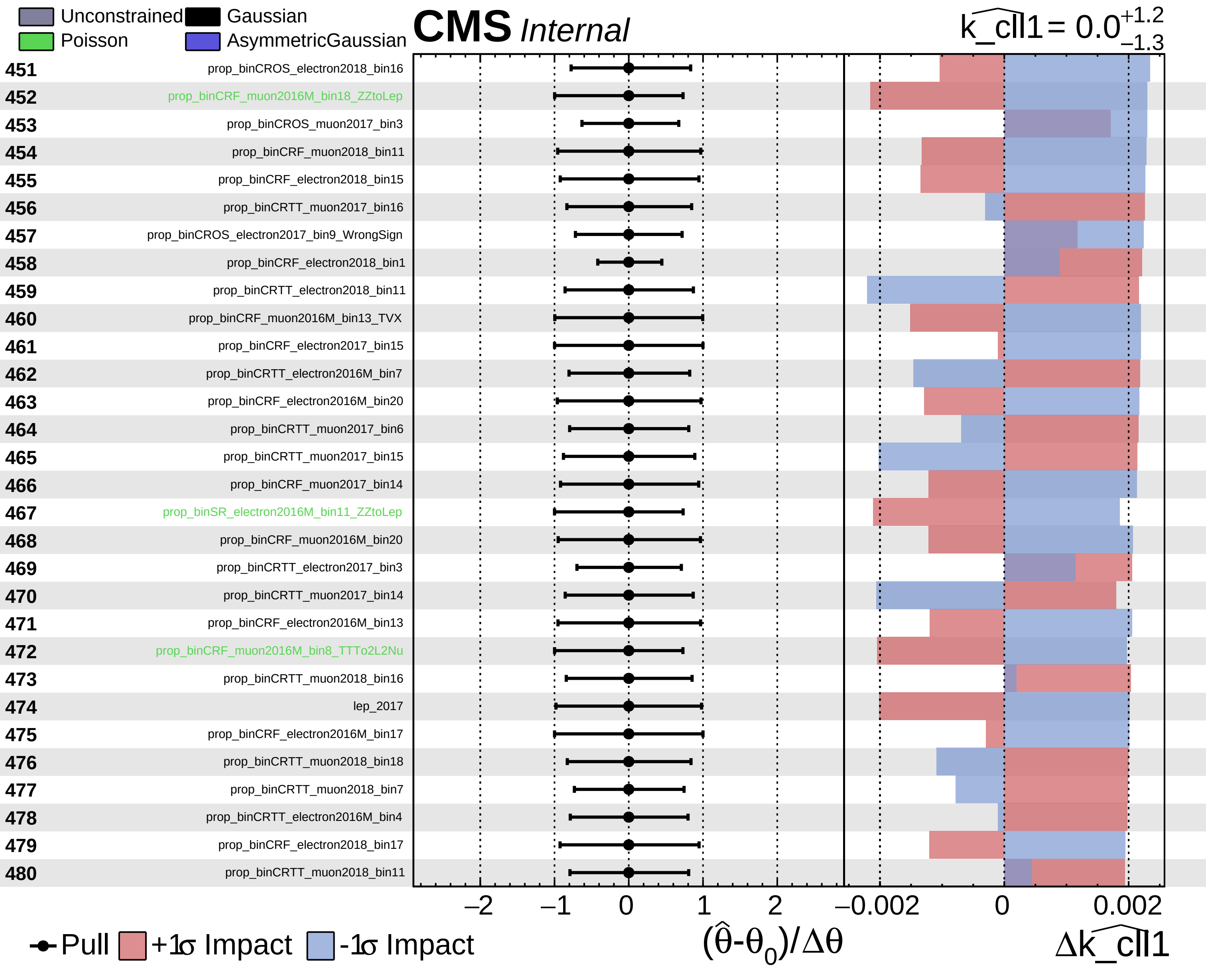


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
361 prop_binCROS_electron2018_bin4
362 prop_binCRF_muon2017_bin9_TTTo2L2Nu
363 prop_binCROS_electron2016M_bin11_WrongSign
364 prop_binCRF_electron2018_bin14_TTTo2L2Nu
365 prop_binCRF_electron2016M_bin8
366 prop_binSR_electron2016M_bin14_sm
367 prop_binCRTT_electron2016M_bin2
368 prop_binCROS_muon2017_bin6_Fake
369 prop_binSR_muon2016M_bin18_ZZtoLep
370 prop_binCRTT_muon2018_bin3
371 prop_binCROS_electron2018_bin5_WrongSign
372 prop_binCRTT_electron2016M_bin1
373 prop_binCRTT_muon2017_bin5
374 prop_binCRTT_muon2016M_bin0
375 prop_binCROS_electron2018_bin17
376 prop_binCRF_muon2017_bin18
377 prop_binCROS_electron2017_bin1
378 prop_binSR_muon2017_bin20_ZZtoLep
379 prop_binCRF_electron2017_bin10
380 prop_binCRTT_muon2017_bin7
381 prop_binCRTT_muon2018_bin19
382 prop_binCROS_muon2016M_bin5_Fake
383 prop_binCROS_muon2017_bin7_Fake
384 prop_binCROS_muon2018_bin8
385 prop_binCRF_electron2017_bin8
386 prop_binCRF_electron2017_bin16_Fake
387 prop_binCROS_electron2017_bin6
388 prop_binCROS_electron2017_bin13_WrongSign
389 prop_binCRF_electron2018_bin19
390 prop_binCROS_muon2018_bin18
−2
−1
0
1
2
−0.004
−0.002
0
0.002
0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cll1}$

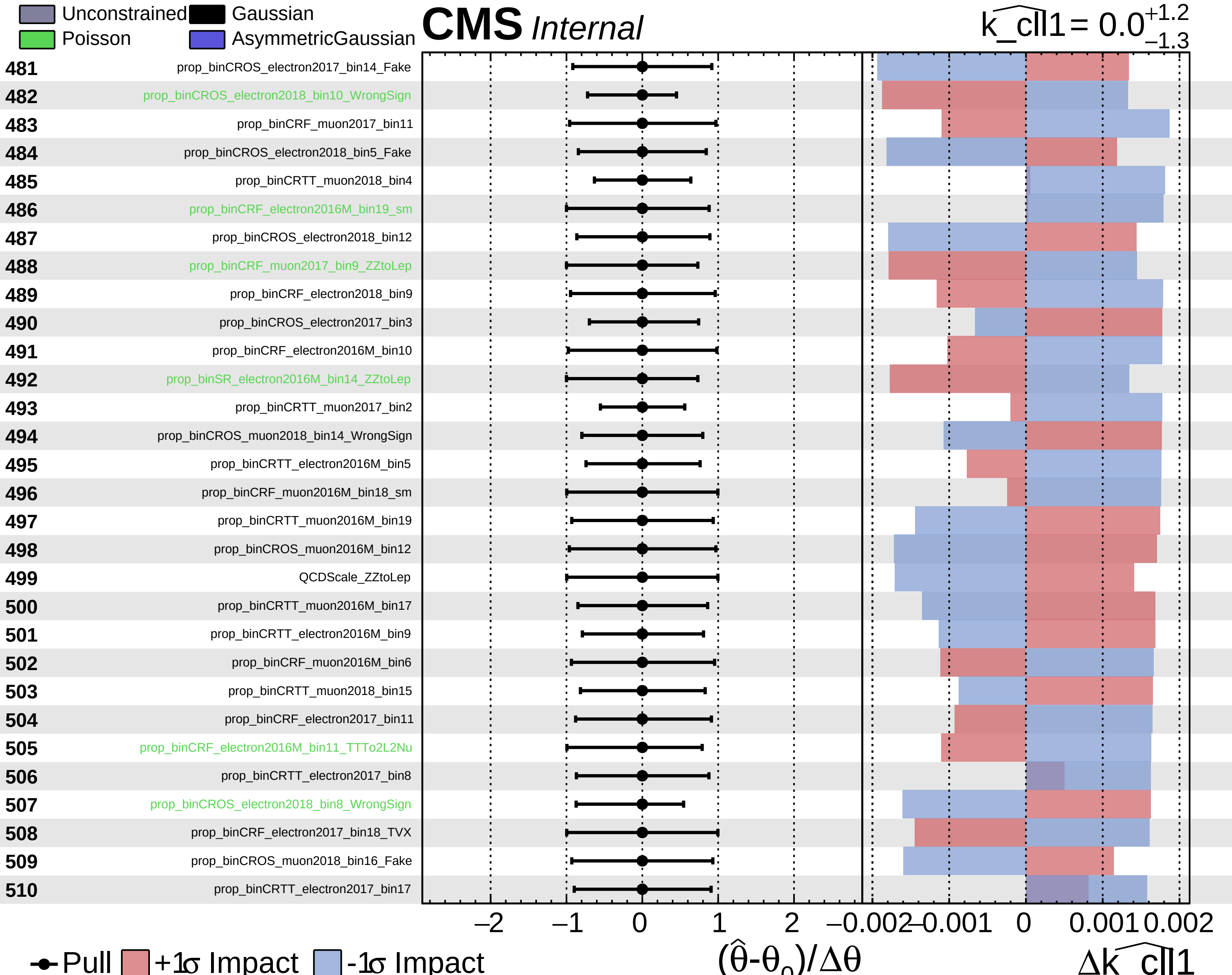

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cll1} = 0.0^{+1.2}_{-1.3}$
391 prop_binCRF_electron2016M_bin5
392 prop_binCRF_electron2017_bin14
393 prop_binCRF_muon2017_bin12
394 prop_binCRTT_muon2016M_bin6
395 prop_binCRF_muon2017_bin10
396 prop_binCRF_muon2017_bin15
397 prop_binCRTT_muon2017_bin20
398 prop_binCROS_muon2017_bin14_WrongSign
399 prop_binSR_electron2017_bin14_sm
400 prop_binSR_electron2017_bin14_TTTo2L2Nu
401 prop_binCRTT_electron2016M_bin3
402 prop_binCROS_electron2017_bin10_Fake
403 prop_binCRF_muon2016M_bin18_Fake
404 prop_binCRF_muon2017_bin19
405 prop_binCROS_muon2017_bin17
406 prop_binCROS_muon2018_bin15_WrongSign
407 prop_binCROS_electron2018_bin13_WrongSign
408 prop_binCROS_electron2018_bin7
409 prop_binCROS_electron2018_bin3
410 prop_binCROS_muon2018_bin10
411 prop_binCROS_electron2018_bin6_WrongSign
412 prop_binCRF_muon2016M_bin16
413 prop_binCRF_muon2016M_bin1
414 prop_binCROS_electron2017_bin8_WrongSign
415 prop_binSR_electron2016M_bin14_Fake
416 prop_binCRF_electron2018_bin8
417 prop_binCRF_electron2016M_bin12_WrongSign
418 prop_binCROS_muon2018_bin20
419 prop_binCRTT_electron2018_bin10
420 prop_binCRTT_electron2016M_bin8
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cll1}$



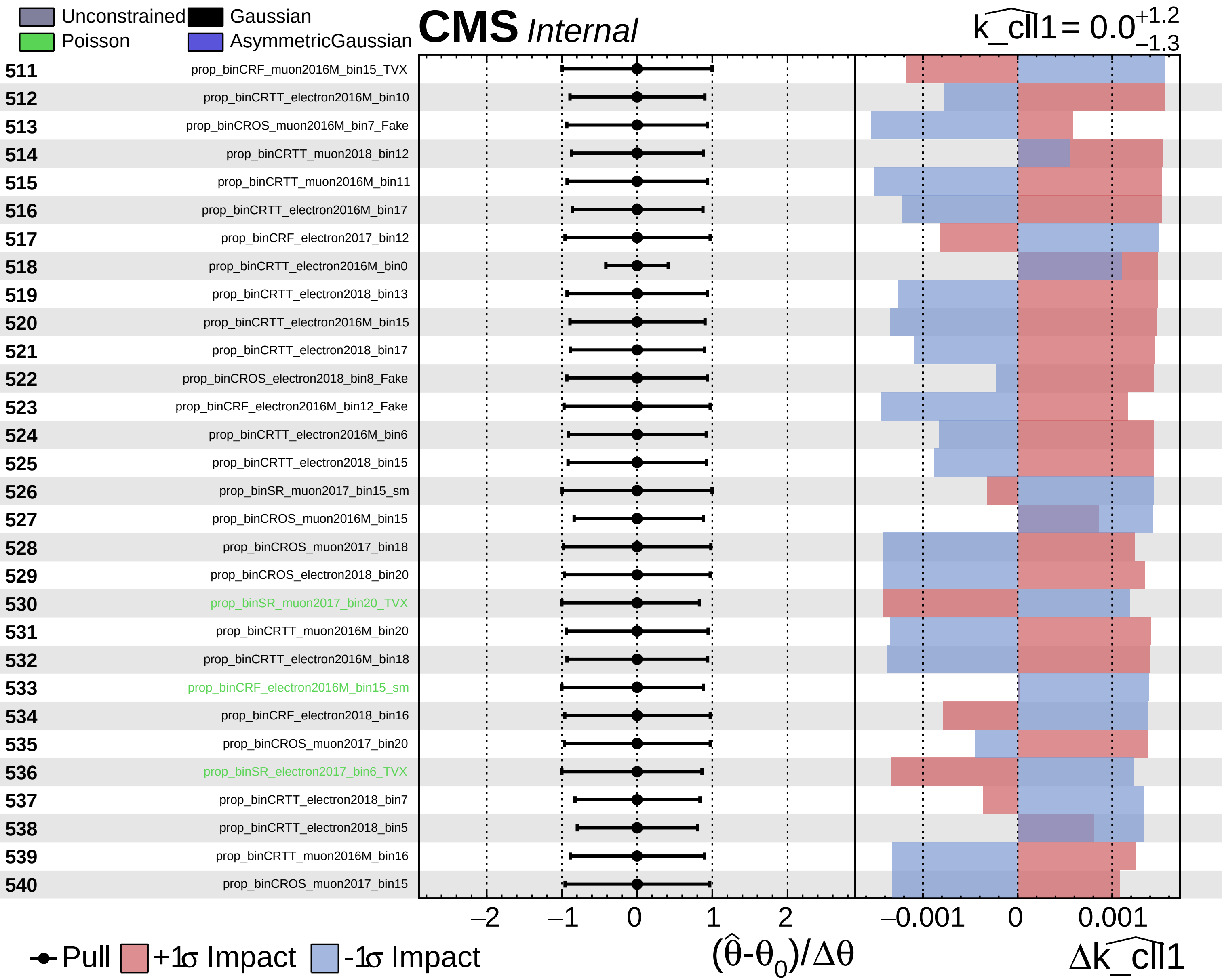
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
421 prop_binCRTT_muon2017_bin11
422 prop_binCROS_muon2018_bin9
423 prop_binCRTT_electron2018_bin9
424 prop_binCRTT_electron2018_bin6
425 prop_binCRTT_muon2017_bin13
426 prop_binCROS_muon2017_bin2
427 prop_binSR_electron2017_bin14_ZZtoLep
428 prop_binCRTT_muon2018_bin6
429 prop_binCRTT_electron2018_bin3
430 prop_binCRF_muon2017_bin6
431 prop_binCROS_muon2017_bin8_WrongSign
432 prop_binCRTT_muon2017_bin18
433 prop_binCRTT_muon2016M_bin9
434 prop_binCRF_muon2017_bin16
435 prop_binCRF_muon2016M_bin17_sm
436 prop_binCRF_muon2016M_bin14
437 prop_binCRF_electron2018_bin13
438 prop_binCRF_electron2018_bin11
439 prop_binCRTT_muon2018_bin10
440 prop_binCROS_muon2017_bin11
441 prop_binCRF_electron2016M_bin6
442 prop_binCRF_muon2016M_bin19_ZZtoLep
443 prop_binCROS_muon2018_bin17
444 prop_binCROS_electron2017_bin2
445 prop_binCROS_muon2017_bin19_WrongSign
446 prop_binCRF_electron2017_bin13_sm
447 prop_binCROS_muon2017_bin12_WrongSign
448 prop_binCRF_muon2018_bin15
449 prop_binCRF_muon2018_bin14
450 prop_binSR_muon2018_bin11_ZZtoLep
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cll1}$



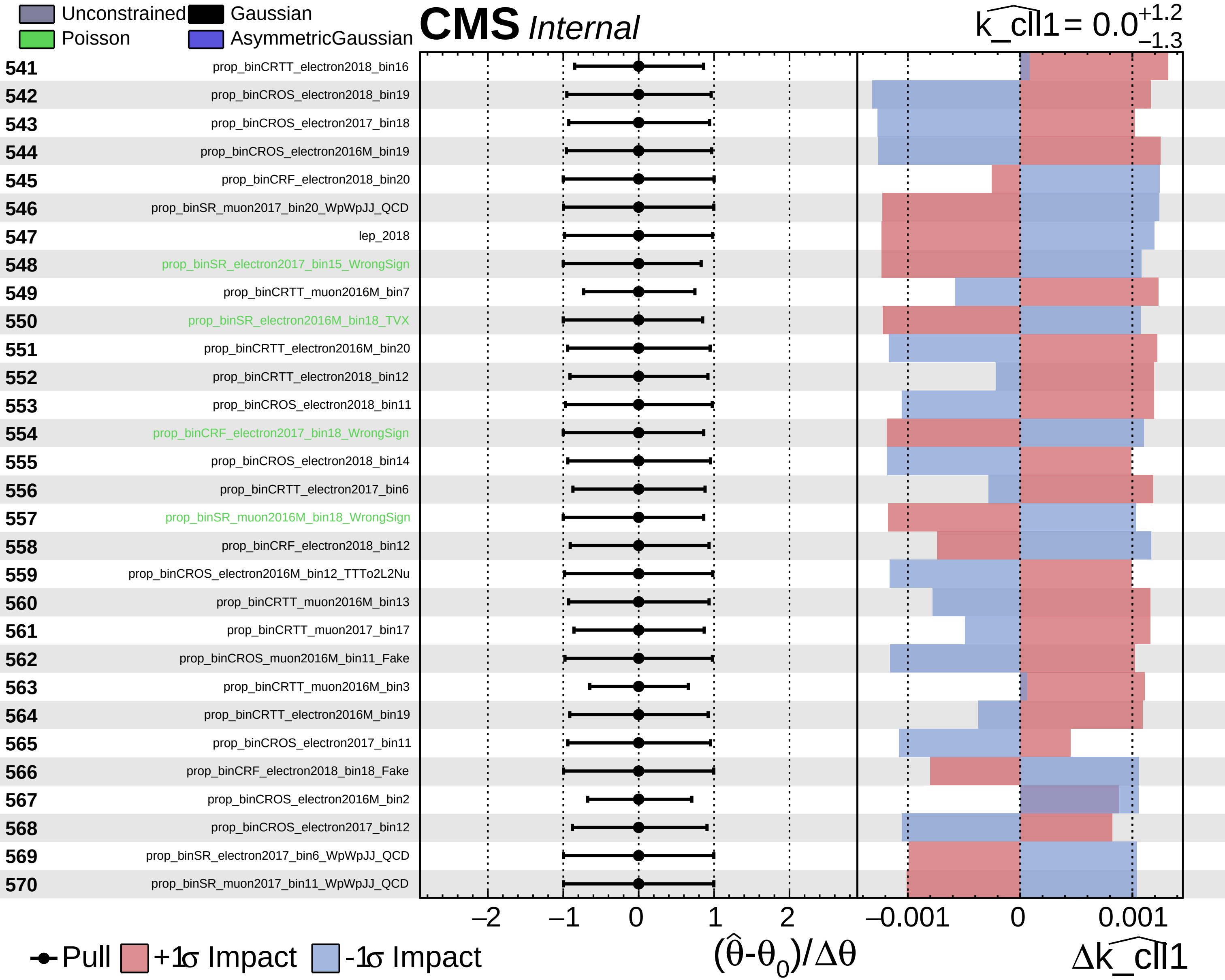
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
451 prop_binCROS_electron2018_bin16
452 prop_binCRF_muon2016M_bin18_ZZtoLep
453 prop_binCROS_muon2017_bin3
454 prop_binCRF_muon2018_bin11
455 prop_binCRF_electron2018_bin15
456 prop_binCRTT_muon2017_bin16
457 prop_binCROS_electron2017_bin9_WrongSign
458 prop_binCRF_electron2018_bin1
459 prop_binCRTT_electron2018_bin11
460 prop_binCRF_muon2016M_bin13_TVX
461 prop_binCRF_electron2017_bin15
462 prop_binCRTT_electron2016M_bin7
463 prop_binCRF_electron2016M_bin20
464 prop_binCRTT_muon2017_bin6
465 prop_binCRTT_muon2017_bin15
466 prop_binCRF_muon2017_bin14
467 prop_binSR_electron2016M_bin11_ZZtoLep
468 prop_binCRF_muon2016M_bin20
469 prop_binCRTT_electron2017_bin3
470 prop_binCRTT_muon2017_bin14
471 prop_binCRF_electron2016M_bin13
472 prop_binCRF_muon2016M_bin8_TTTo2L2Nu
473 prop_binCRTT_muon2018_bin16
474 lep_2017
475 prop_binCRF_electron2016M_bin17
476 prop_binCRTT_muon2018_bin18
477 prop_binCRTT_muon2018_bin7
478 prop_binCRTT_electron2016M_bin4
479 prop_binCRF_electron2018_bin17
480 prop_binCRTT_muon2018_bin11
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cll1

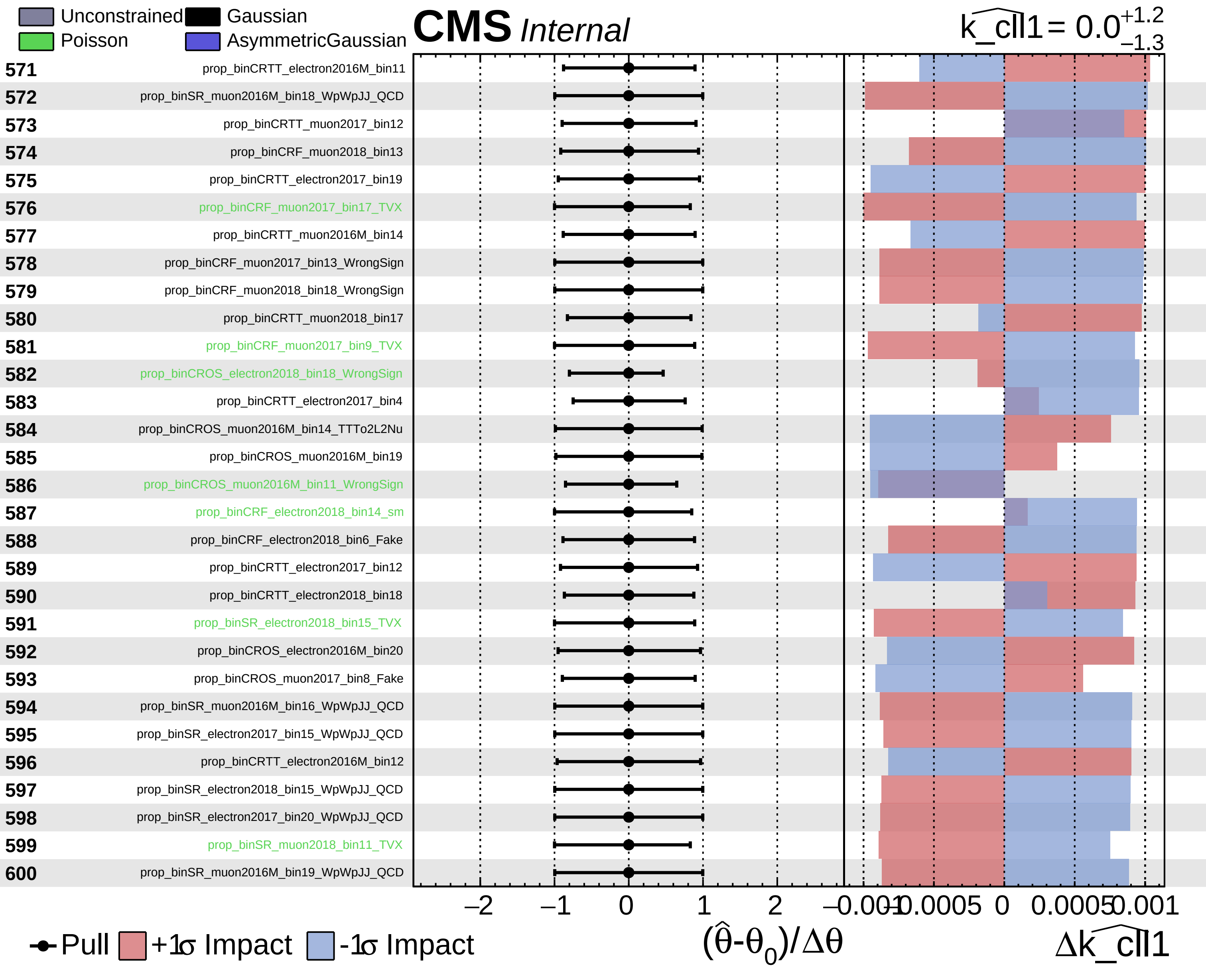


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cll1} = 0.0^{+1.2}_{-1.3}$
481 prop_binCROS_electron2017_bin14_Fake
482 prop_binCROS_electron2018_bin10_WrongSign
483 prop_binCRF_muon2017_bin11
484 prop_binCROS_electron2018_bin5_Fake
485 prop_binCRTT_muon2018_bin4
486 prop_binCRF_electron2016M_bin19_sm
487 prop_binCROS_electron2018_bin12
488 prop_binCRF_muon2017_bin9_ZZtoLep
489 prop_binCRF_electron2018_bin9
490 prop_binCROS_electron2017_bin3
491 prop_binCRF_electron2016M_bin10
492 prop_binSR_electron2016M_bin14_ZZtoLep
493 prop_binCRTT_muon2017_bin2
494 prop_binCROS_muon2018_bin14_WrongSign
495 prop_binCRTT_electron2016M_bin5
496 prop_binCRF_muon2016M_bin18_sm
497 prop_binCRTT_muon2016M_bin19
498 prop_binCROS_muon2016M_bin12
499 QCDScale_ZZtoLep
500 prop_binCRTT_muon2016M_bin17
501 prop_binCRTT_electron2016M_bin9
502 prop_binCRF_muon2016M_bin6
503 prop_binCRTT_muon2018_bin15
504 prop_binCRF_electron2017_bin11
505 prop_binCRF_electron2016M_bin11_TTTo2L2Nu
506 prop_binCRTT_electron2017_bin8
507 prop_binCROS_electron2018_bin8_WrongSign
508 prop_binCRF_electron2017_bin18_TVX
509 prop_binCROS_muon2018_bin16_Fake
510 prop_binCRTT_electron2017_bin17
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cll1}$

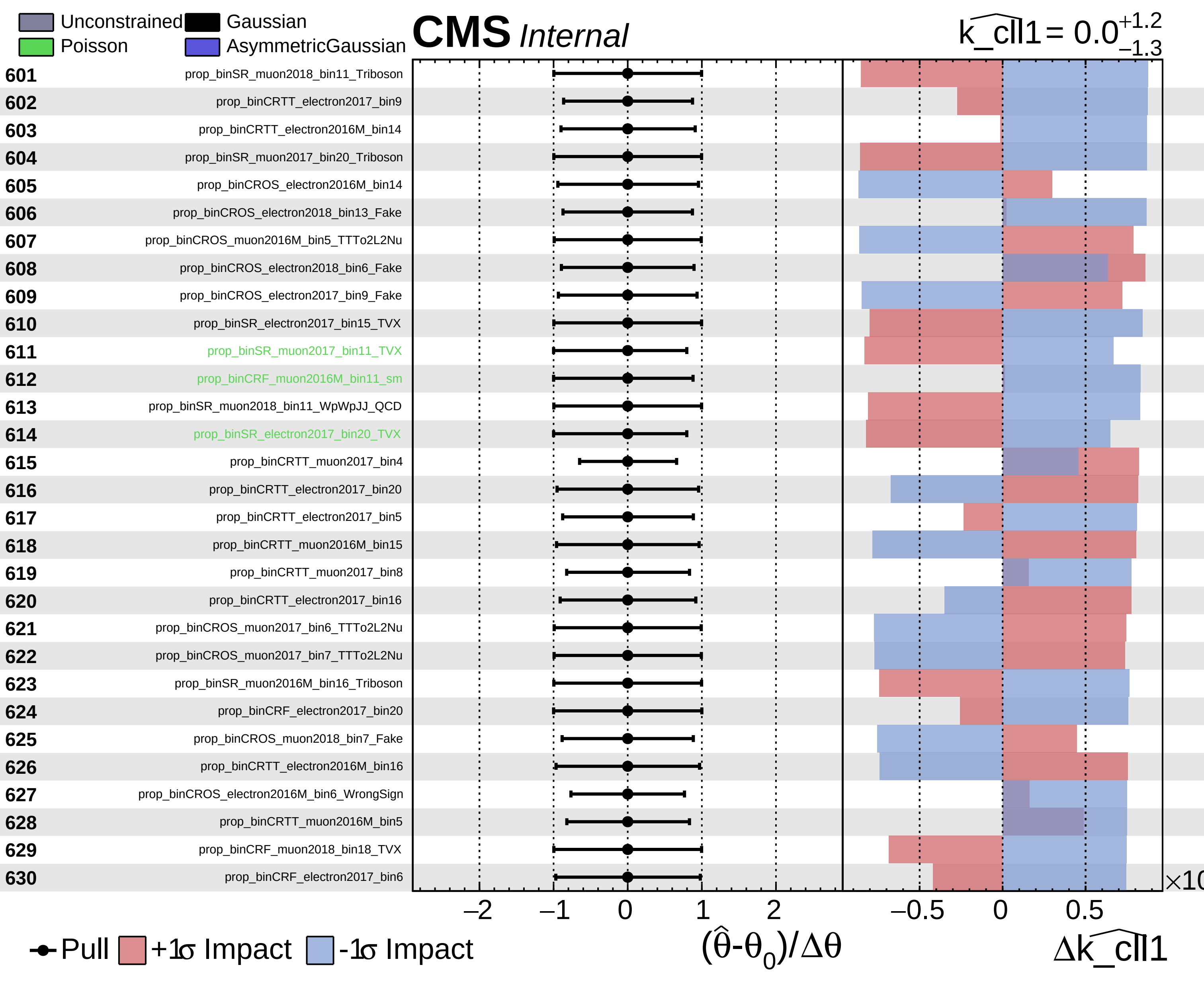


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
511 prop_binCRF_muon2016M_bin15_TVX
512 prop_binCRTT_electron2016M_bin10
513 prop_binCROS_muon2016M_bin7_Fake
514 prop_binCRTT_muon2018_bin12
515 prop_binCRTT_muon2016M_bin11
516 prop_binCRTT_electron2016M_bin17
517 prop_binCRF_electron2017_bin12
518 prop_binCRTT_electron2016M_bin0
519 prop_binCRTT_electron2018_bin13
520 prop_binCRTT_electron2016M_bin15
521 prop_binCRTT_electron2018_bin17
522 prop_binCROS_electron2018_bin8_Fake
523 prop_binCRF_electron2016M_bin12_Fake
524 prop_binCRTT_electron2016M_bin6
525 prop_binCRTT_electron2018_bin15
526 prop_binSR_muon2017_bin15_sm
527 prop_binCROS_muon2016M_bin15
528 prop_binCROS_muon2017_bin18
529 prop_binCROS_electron2018_bin20
530 prop_binSR_muon2017_bin20_TVX
531 prop_binCRTT_muon2016M_bin20
532 prop_binCRTT_electron2016M_bin18
533 prop_binCRF_electron2016M_bin15_sm
534 prop_binCRF_electron2018_bin16
535 prop_binCROS_muon2017_bin20
536 prop_binSR_electron2017_bin6_TVX
537 prop_binCRTT_electron2018_bin7
538 prop_binCRTT_electron2018_bin5
539 prop_binCRTT_muon2016M_bin16
540 prop_binCROS_muon2017_bin15
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cll1

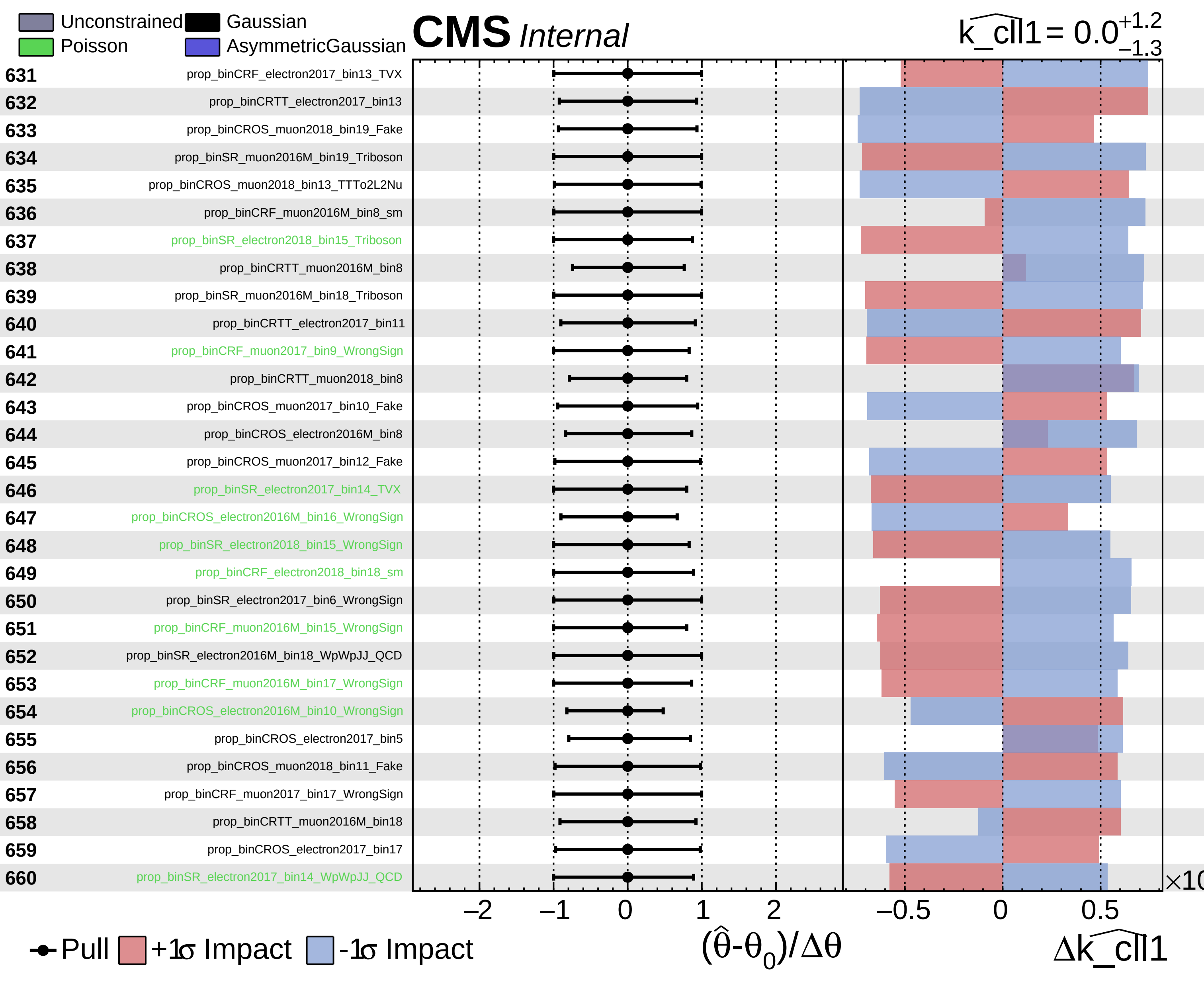

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
541 prop_binCRTT_electron2018_bin16
542 prop_binCROS_electron2018_bin19
543 prop_binCROS_electron2017_bin18
544 prop_binCROS_electron2016M_bin19
545 prop_binCRF_electron2018_bin20
546 prop_binSR_muon2017_bin20_WpWpJJ_QCD
547 lep_2018
548 prop_binSR_electron2017_bin15_WrongSign
549 prop_binCRTT_muon2016M_bin7
550 prop_binSR_electron2016M_bin18_TVX
551 prop_binCRTT_electron2016M_bin20
552 prop_binCRTT_electron2018_bin12
553 prop_binCROS_electron2018_bin11
554 prop_binCRF_electron2017_bin18_WrongSign
555 prop_binCROS_electron2018_bin14
556 prop_binCRTT_electron2017_bin6
557 prop_binSR_muon2016M_bin18_WrongSign
558 prop_binCRF_electron2018_bin12
559 prop_binCROS_electron2016M_bin12_TTTo2L2Nu
560 prop_binCRTT_muon2016M_bin13
561 prop_binCRTT_muon2017_bin17
562 prop_binCROS_muon2016M_bin11_Fake
563 prop_binCRTT_muon2016M_bin3
564 prop_binCRTT_electron2016M_bin19
565 prop_binCROS_electron2017_bin11
566 prop_binCRF_electron2018_bin18_Fake
567 prop_binCROS_electron2016M_bin2
568 prop_binCROS_electron2017_bin12
569 prop_binSR_electron2017_bin6_WpWpJJ_QCD
570 prop_binSR_muon2017_bin11_WpWpJJ_QCD
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cll1



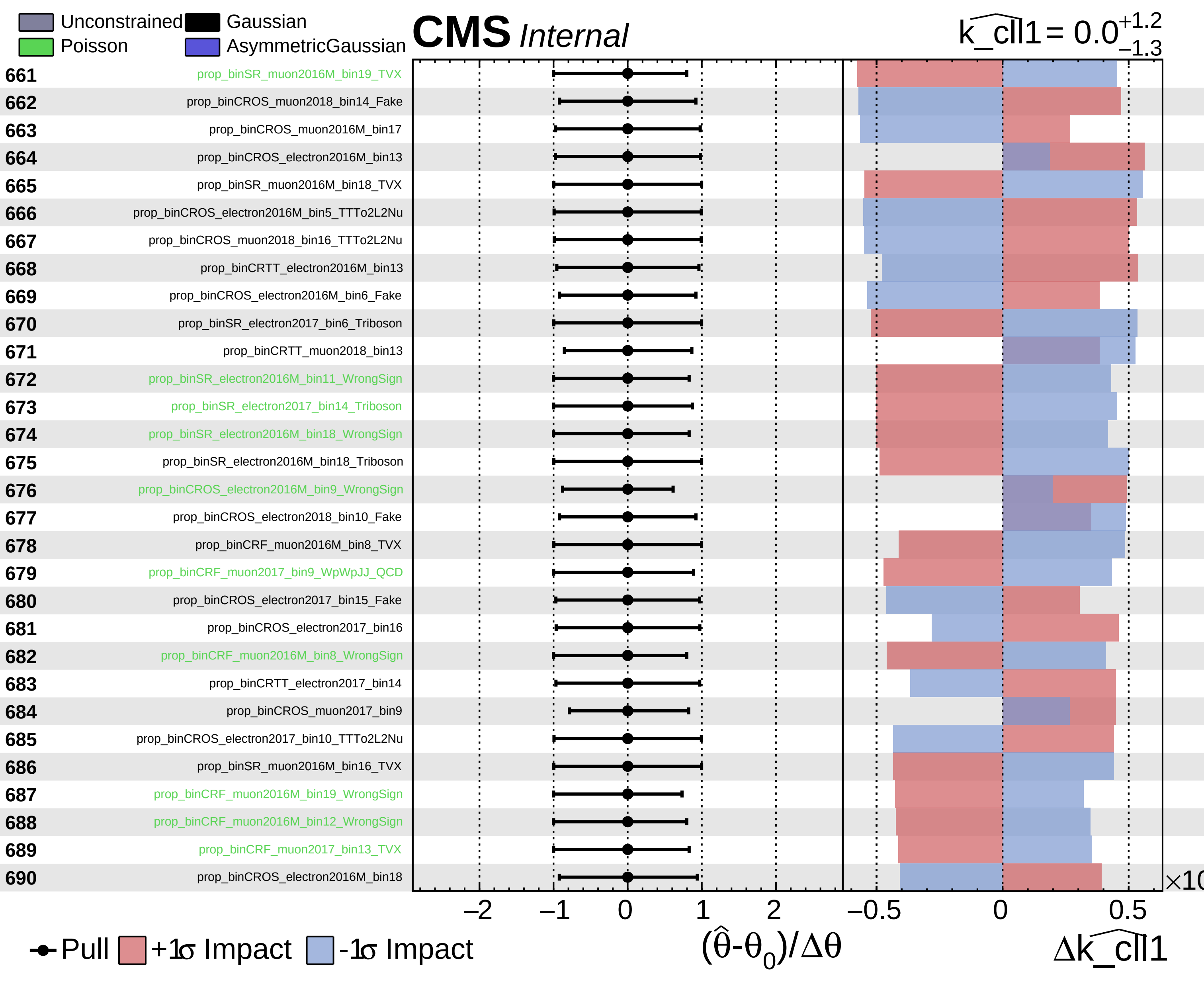
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
571 prop_binCRTT_electron2016M_bin11
572 prop_binSR_muon2016M_bin18_WpWpJJ_QCD
573 prop_binCRTT_muon2017_bin12
574 prop_binCRF_muon2018_bin13
575 prop_binCRTT_electron2017_bin19
576 prop_binCRF_muon2017_bin17_TVX
577 prop_binCRTT_muon2016M_bin14
578 prop_binCRF_muon2017_bin13_WrongSign
579 prop_binCRF_muon2018_bin18_WrongSign
580 prop_binCRTT_muon2018_bin17
581 prop_binCRF_muon2017_bin9_TVX
582 prop_binCROS_electron2018_bin18_WrongSign
583 prop_binCRTT_electron2017_bin4
584 prop_binCROS_muon2016M_bin14_TTTo2L2Nu
585 prop_binCROS_muon2016M_bin19
586 prop_binCROS_muon2016M_bin11_WrongSign
587 prop_binCRF_electron2018_bin14_sm
588 prop_binCRF_electron2018_bin6_Fake
589 prop_binCRTT_electron2017_bin12
590 prop_binCRTT_electron2018_bin18
591 prop_binSR_electron2018_bin15_TVX
592 prop_binCROS_electron2016M_bin20
593 prop_binCROS_muon2017_bin8_Fake
594 prop_binSR_muon2016M_bin16_WpWpJJ_QCD
595 prop_binSR_electron2017_bin15_WpWpJJ_QCD
596 prop_binCRTT_electron2016M_bin12
597 prop_binSR_electron2018_bin15_WpWpJJ_QCD
598 prop_binSR_electron2017_bin20_WpWpJJ_QCD
599 prop_binSR_muon2018_bin11_TVX
600 prop_binSR_muon2016M_bin19_WpWpJJ_QCD
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cll1}$



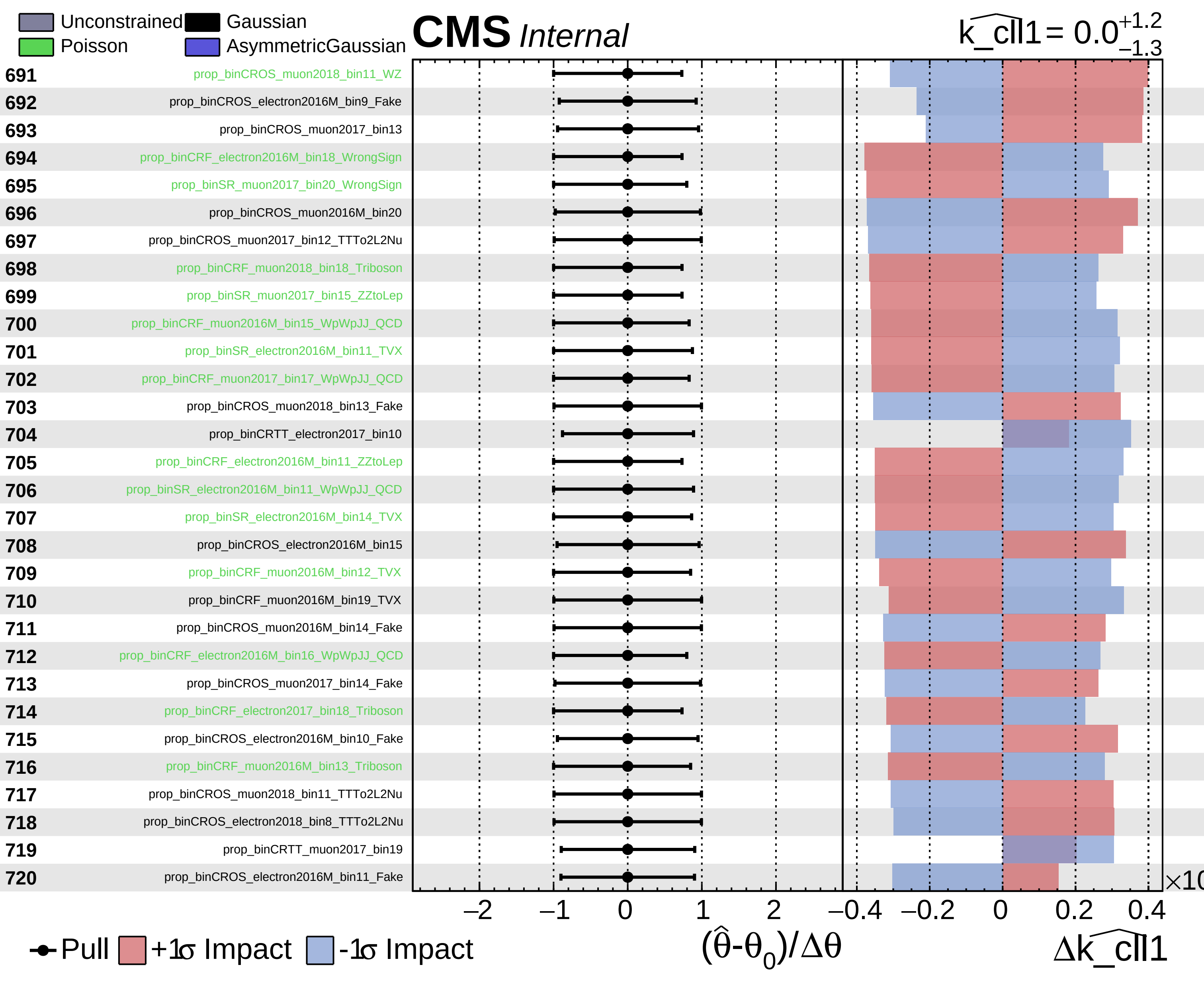
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cll1} = 0.0^{+1.2}_{-1.3}$
601 prop_binSR_muon2018_bin11_Triboson
602 prop_binCRTT_electron2017_bin9
603 prop_binCRTT_electron2016M_bin14
604 prop_binSR_muon2017_bin20_Triboson
605 prop_binCROS_electron2016M_bin14
606 prop_binCROS_electron2018_bin13_Fake
607 prop_binCROS_muon2016M_bin5_TTTo2L2Nu
608 prop_binCROS_electron2018_bin6_Fake
609 prop_binCROS_electron2017_bin9_Fake
610 prop_binSR_electron2017_bin15_TVX
611 prop_binSR_muon2017_bin11_TVX
612 prop_binCRF_muon2016M_bin11_sm
613 prop_binSR_muon2018_bin11_WpWpJJ_QCD
614 prop_binSR_electron2017_bin20_TVX
615 prop_binCRTT_muon2017_bin4
616 prop_binCRTT_electron2017_bin20
617 prop_binCRTT_electron2017_bin5
618 prop_binCRTT_muon2016M_bin15
619 prop_binCRTT_muon2017_bin8
620 prop_binCRTT_electron2017_bin16
621 prop_binCROS_muon2017_bin6_TTTo2L2Nu
622 prop_binCROS_muon2017_bin7_TTTo2L2Nu
623 prop_binSR_muon2016M_bin16_Triboson
624 prop_binCRF_electron2017_bin20
625 prop_binCROS_muon2018_bin7_Fake
626 prop_binCRTT_electron2016M_bin16
627 prop_binCROS_electron2016M_bin6_WrongSign
628 prop_binCRTT_muon2016M_bin5
629 prop_binCRF_muon2018_bin18_TVX
630 prop_binCRF_electron2017_bin6
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cll1}$



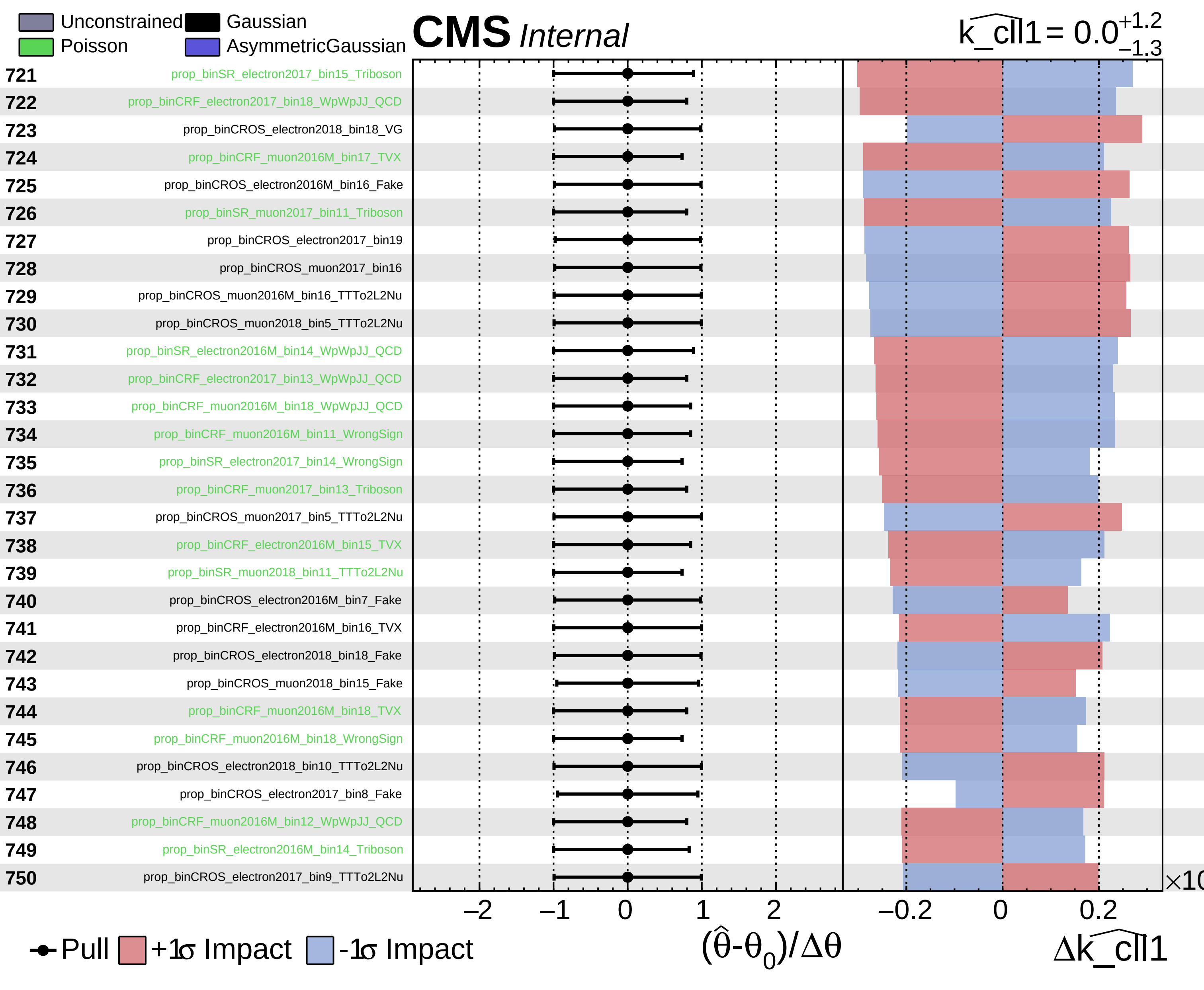
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
631 prop_binCRF_electron2017_bin13_TVX
632 prop_binCRTT_electron2017_bin13
633 prop_binCROS_muon2018_bin19_Fake
634 prop_binSR_muon2016M_bin19_Triboson
635 prop_binCROS_muon2018_bin13_TTTo2L2Nu
636 prop_binCRF_muon2016M_bin8_sm
637 prop_binSR_electron2018_bin15_Triboson
638 prop_binCRTT_muon2016M_bin8
639 prop_binSR_muon2016M_bin18_Triboson
640 prop_binCRTT_electron2017_bin11
641 prop_binCRF_muon2017_bin9_WrongSign
642 prop_binCRTT_muon2018_bin8
643 prop_binCROS_muon2017_bin10_Fake
644 prop_binCROS_electron2016M_bin8
645 prop_binCROS_muon2017_bin12_Fake
646 prop_binSR_electron2017_bin14_TVX
647 prop_binCROS_electron2016M_bin16_WrongSign
648 prop_binSR_electron2018_bin15_WrongSign
649 prop_binCRF_electron2018_bin18_sm
650 prop_binSR_electron2017_bin6_WrongSign
651 prop_binCRF_muon2016M_bin15_WrongSign
652 prop_binSR_electron2016M_bin18_WpWpJJ_QCD
653 prop_binCRF_muon2016M_bin17_WrongSign
654 prop_binCROS_electron2016M_bin10_WrongSign
655 prop_binCROS_electron2017_bin5
656 prop_binCROS_muon2018_bin11_Fake
657 prop_binCRF_muon2017_bin17_WrongSign
658 prop_binCRTT_muon2016M_bin18
659 prop_binCROS_electron2017_bin17
660 prop_binSR_electron2017_bin14_WpWpJJ_QCD
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cll1



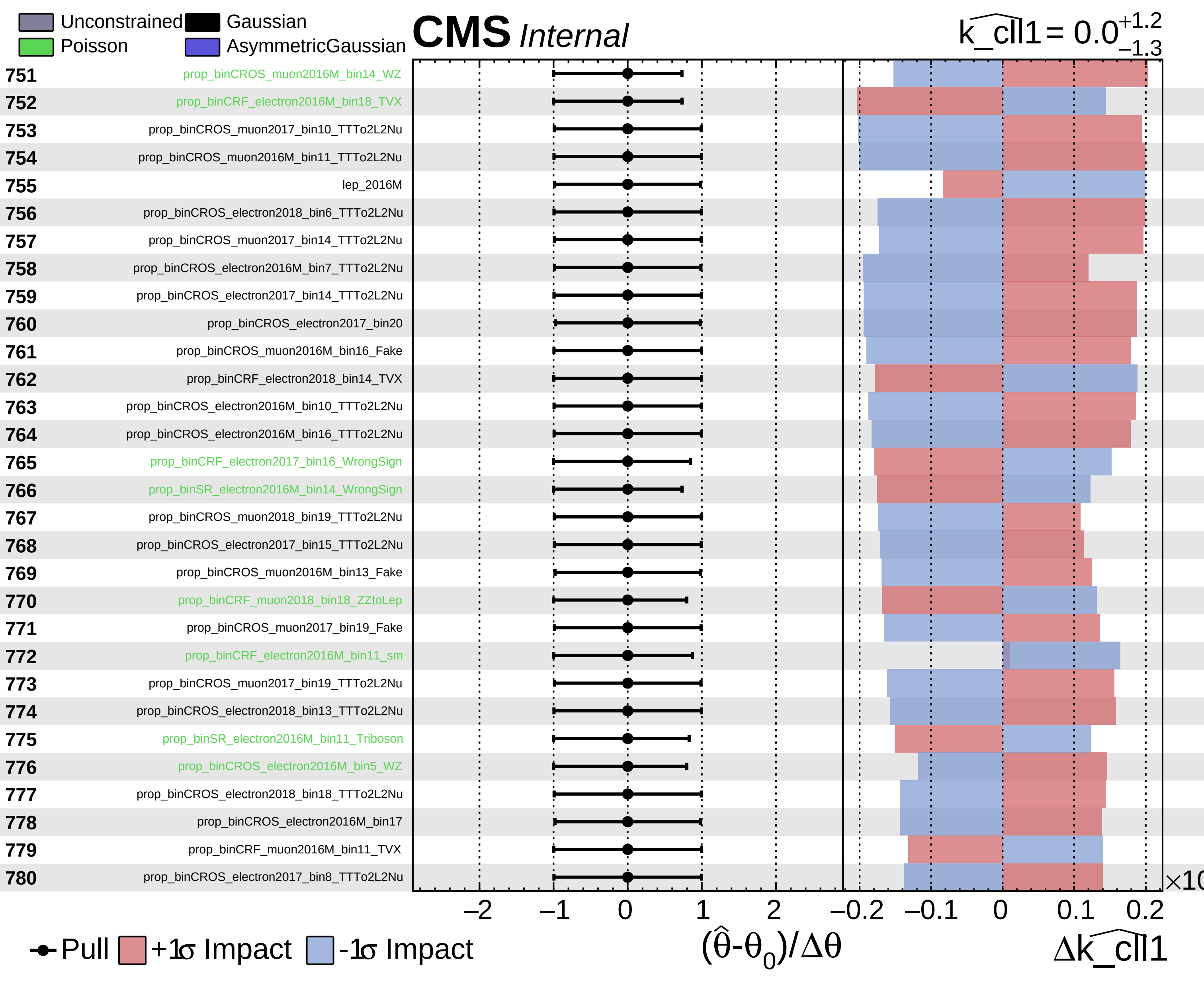
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
661 prop_binSR_muon2016M_bin19_TVX
662 prop_binCROS_muon2018_bin14_Fake
663 prop_binCROS_muon2016M_bin17
664 prop_binCROS_electron2016M_bin13
665 prop_binSR_muon2016M_bin18_TVX
666 prop_binCROS_electron2016M_bin5_TTTo2L2Nu
667 prop_binCROS_muon2018_bin16_TTTo2L2Nu
668 prop_binCRTT_electron2016M_bin13
669 prop_binCROS_electron2016M_bin6_Fake
670 prop_binSR_electron2017_bin6_Triboson
671 prop_binCRTT_muon2018_bin13
672 prop_binSR_electron2016M_bin11_WrongSign
673 prop_binSR_electron2017_bin14_Triboson
674 prop_binSR_electron2016M_bin18_WrongSign
675 prop_binSR_electron2016M_bin18_Triboson
676 prop_binCROS_electron2016M_bin9_WrongSign
677 prop_binCROS_electron2018_bin10_Fake
678 prop_binCRF_muon2016M_bin8_TVX
679 prop_binCRF_muon2017_bin9_WpWpJJ_QCD
680 prop_binCROS_electron2017_bin15_Fake
681 prop_binCROS_electron2017_bin16
682 prop_binCRF_muon2016M_bin8_WrongSign
683 prop_binCRTT_electron2017_bin14
684 prop_binCROS_muon2017_bin9
685 prop_binCROS_electron2017_bin10_TTTo2L2Nu
686 prop_binSR_muon2016M_bin16_TVX
687 prop_binCRF_muon2016M_bin19_WrongSign
688 prop_binCRF_muon2016M_bin12_WrongSign
689 prop_binCRF_muon2017_bin13_TVX
690 prop_binCROS_electron2016M_bin18
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cll1



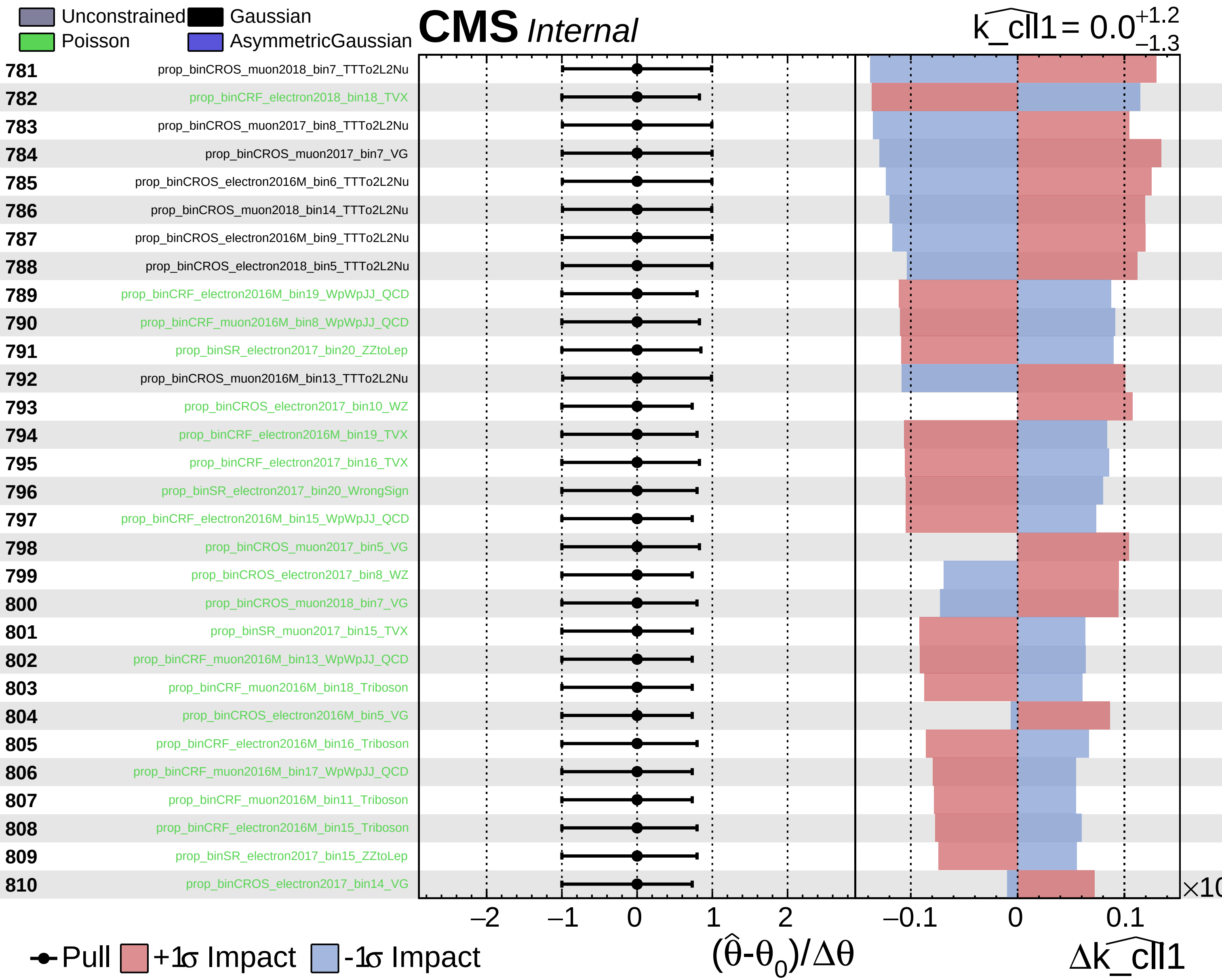
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cll1} = 0.0^{+1.2}_{-1.3}$
691 prop_binCROS_muon2018_bin11_WZ
692 prop_binCROS_electron2016M_bin9_Fake
693 prop_binCROS_muon2017_bin13
694 prop_binCRF_electron2016M_bin18_WrongSign
695 prop_binSR_muon2017_bin20_WrongSign
696 prop_binCROS_muon2016M_bin20
697 prop_binCROS_muon2017_bin12_TTTo2L2Nu
698 prop_binCRF_muon2018_bin18_Triboson
699 prop_binSR_muon2017_bin15_ZZtoLep
700 prop_binCRF_muon2016M_bin15_WpWpJJ_QCD
701 prop_binSR_electron2016M_bin11_TVX
702 prop_binCRF_muon2017_bin17_WpWpJJ_QCD
703 prop_binCROS_muon2018_bin13_Fake
704 prop_binCRTT_electron2017_bin10
705 prop_binCRF_electron2016M_bin11_ZZtoLep
706 prop_binSR_electron2016M_bin11_WpWpJJ_QCD
707 prop_binSR_electron2016M_bin14_TVX
708 prop_binCROS_electron2016M_bin15
709 prop_binCRF_muon2016M_bin12_TVX
710 prop_binCRF_muon2016M_bin19_TVX
711 prop_binCROS_muon2016M_bin14_Fake
712 prop_binCRF_electron2016M_bin16_WpWpJJ_QCD
713 prop_binCROS_muon2017_bin14_Fake
714 prop_binCRF_electron2017_bin18_Triboson
715 prop_binCROS_electron2016M_bin10_Fake
716 prop_binCRF_muon2016M_bin13_Triboson
717 prop_binCROS_muon2018_bin11_TTTo2L2Nu
718 prop_binCROS_electron2018_bin8_TTTo2L2Nu
719 prop_binCRTT_muon2017_bin19
720 prop_binCROS_electron2016M_bin11_Fake
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cll1}$



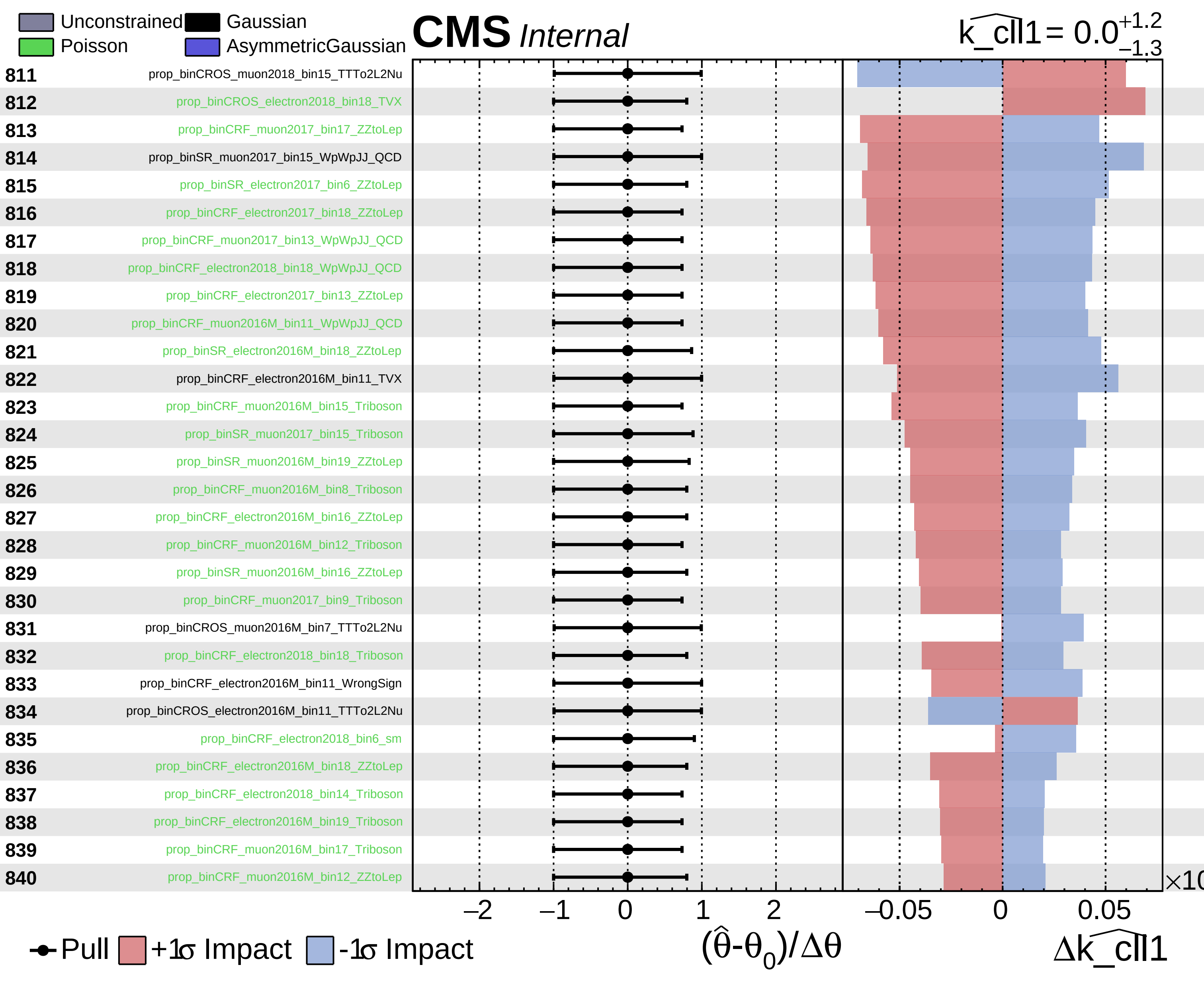
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
721 prop_binSR_electron2017_bin15_Triboson
722 prop_binCRF_electron2017_bin18_WpWpJJ_QCD
723 prop_binCROS_electron2018_bin18_VG
724 prop_binCRF_muon2016M_bin17_TVX
725 prop_binCROS_electron2016M_bin16_Fake
726 prop_binSR_muon2017_bin11_Triboson
727 prop_binCROS_electron2017_bin19
728 prop_binCROS_muon2017_bin16
729 prop_binCROS_muon2016M_bin16_TTTo2L2Nu
730 prop_binCROS_muon2018_bin5_TTTo2L2Nu
731 prop_binSR_electron2016M_bin14_WpWpJJ_QCD
732 prop_binCRF_electron2017_bin13_WpWpJJ_QCD
733 prop_binCRF_muon2016M_bin18_WpWpJJ_QCD
734 prop_binCRF_muon2016M_bin11_WrongSign
735 prop_binSR_electron2017_bin14_WrongSign
736 prop_binCRF_muon2017_bin13_Triboson
737 prop_binCROS_muon2017_bin5_TTTo2L2Nu
738 prop_binCRF_electron2016M_bin15_TVX
739 prop_binSR_muon2018_bin11_TTTo2L2Nu
740 prop_binCROS_electron2016M_bin7_Fake
741 prop_binCRF_electron2016M_bin16_TVX
742 prop_binCROS_electron2018_bin18_Fake
743 prop_binCROS_muon2018_bin15_Fake
744 prop_binCRF_muon2016M_bin18_TVX
745 prop_binCRF_muon2016M_bin18_WrongSign
746 prop_binCROS_electron2018_bin10_TTTo2L2Nu
747 prop_binCROS_electron2017_bin8_Fake
748 prop_binCRF_muon2016M_bin12_WpWpJJ_QCD
749 prop_binSR_electron2016M_bin14_Triboson
750 prop_binCROS_electron2017_bin9_TTTo2L2Nu
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cll1



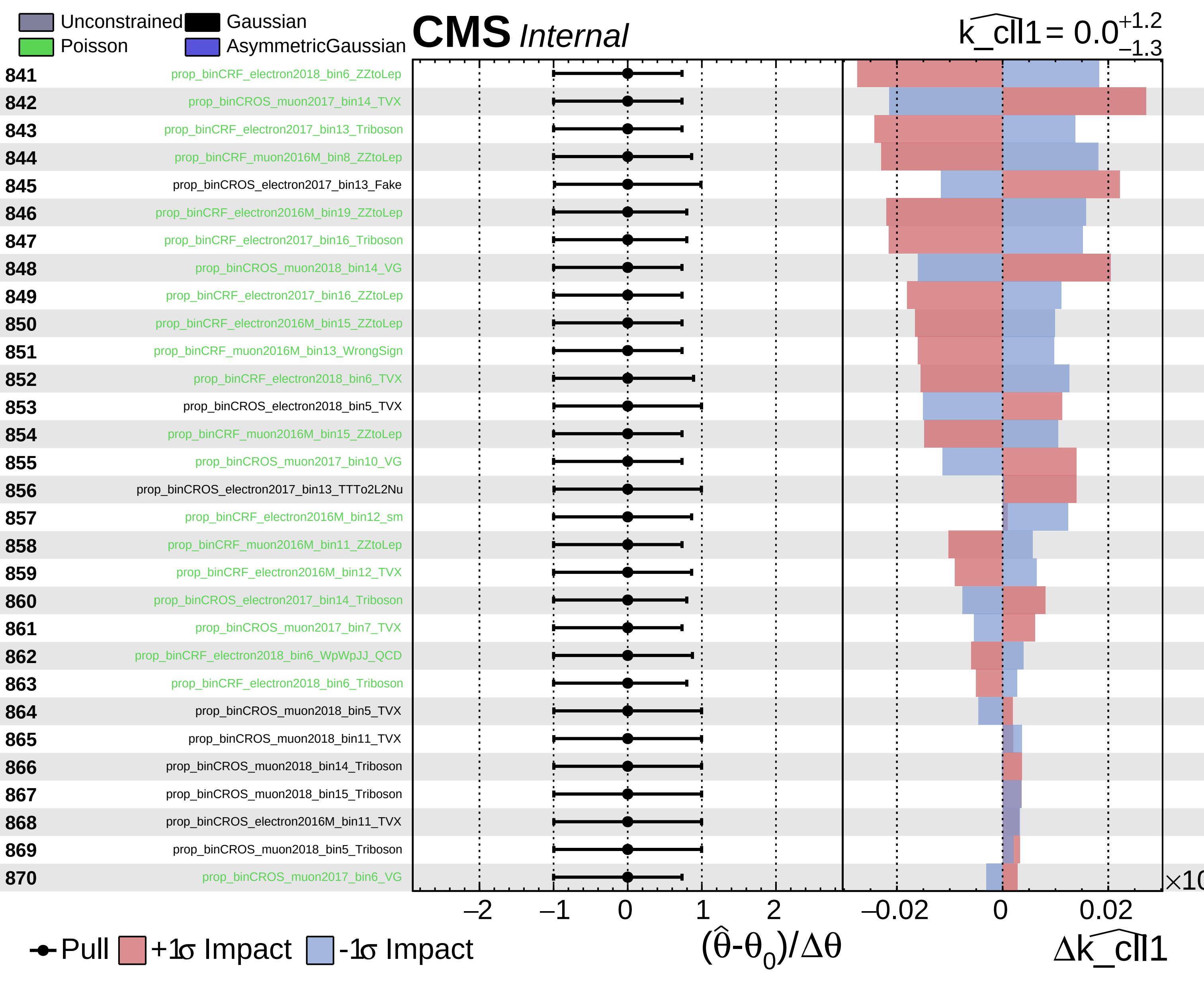
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
751 prop_binCROS_muon2016M_bin14_WZ
752 prop_binCRF_electron2016M_bin18_TVX
753 prop_binCROS_muon2017_bin10_TTTo2L2Nu
754 prop_binCROS_muon2016M_bin11_TTTo2L2Nu
755 lep_2016M
756 prop_binCROS_electron2018_bin6_TTTo2L2Nu
757 prop_binCROS_muon2017_bin14_TTTo2L2Nu
758 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
759 prop_binCROS_electron2017_bin14_TTTo2L2Nu
760 prop_binCROS_electron2017_bin20
761 prop_binCROS_muon2016M_bin16_Fake
762 prop_binCRF_electron2018_bin14_TVX
763 prop_binCROS_electron2016M_bin10_TTTo2L2Nu
764 prop_binCROS_electron2016M_bin16_TTTo2L2Nu
765 prop_binCRF_electron2017_bin16_WrongSign
766 prop_binSR_electron2016M_bin14_WrongSign
767 prop_binCROS_muon2018_bin19_TTTo2L2Nu
768 prop_binCROS_electron2017_bin15_TTTo2L2Nu
769 prop_binCROS_muon2016M_bin13_Fake
770 prop_binCRF_muon2018_bin18_ZZtoLep
771 prop_binCROS_muon2017_bin19_Fake
772 prop_binCRF_electron2016M_bin11_sm
773 prop_binCROS_muon2017_bin19_TTTo2L2Nu
774 prop_binCROS_electron2018_bin13_TTTo2L2Nu
775 prop_binSR_electron2016M_bin11_Triboson
776 prop_binCROS_electron2016M_bin5_WZ
777 prop_binCROS_electron2018_bin18_TTTo2L2Nu
778 prop_binCROS_electron2016M_bin17
779 prop_binCRF_muon2016M_bin11_TVX
780 prop_binCROS_electron2017_bin8_TTTo2L2Nu
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cll1



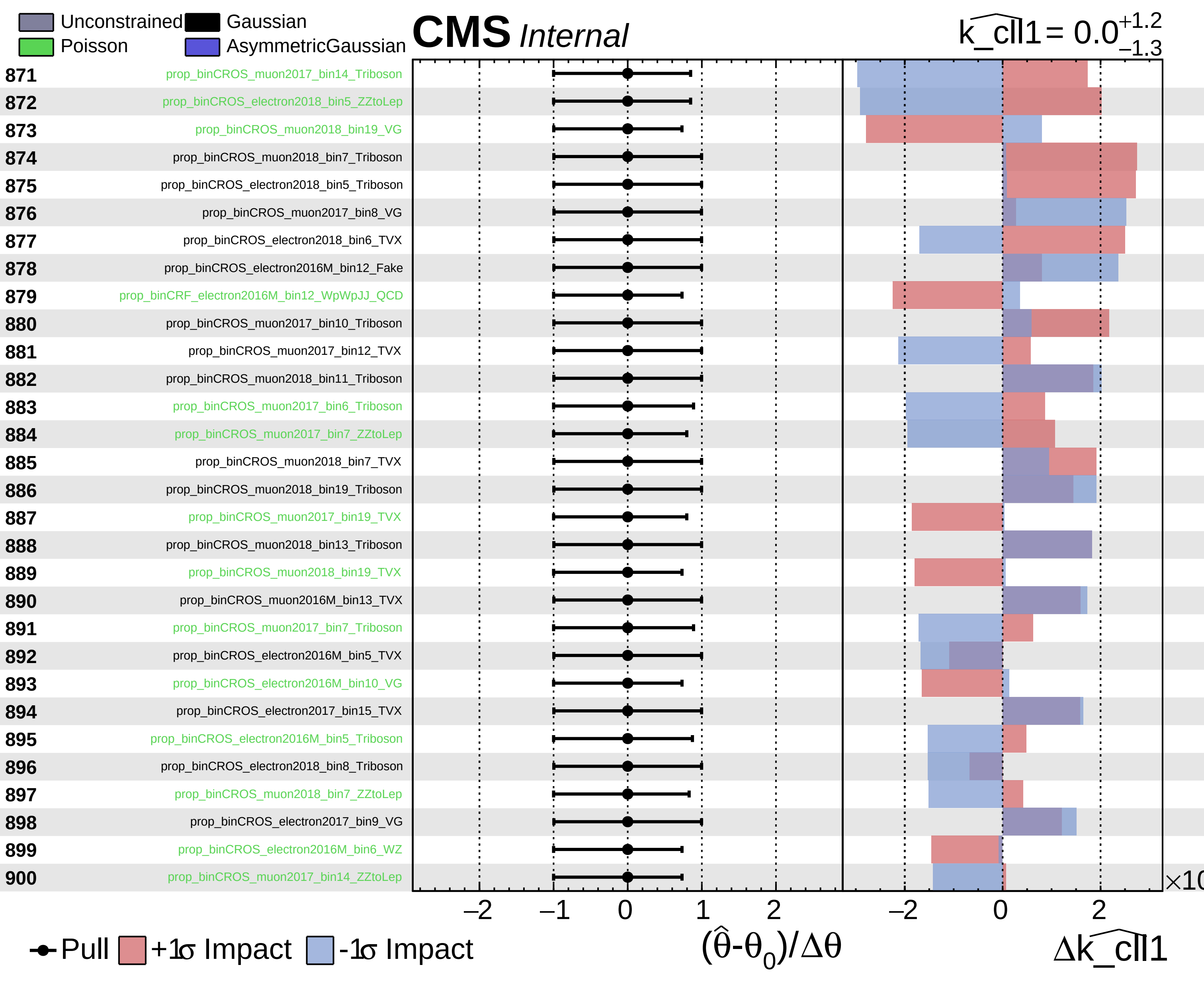
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
781 prop_binCROS_muon2018_bin7_TTTo2L2Nu
782 prop_binCRF_electron2018_bin18_TVX
783 prop_binCROS_muon2017_bin8_TTTo2L2Nu
784 prop_binCROS_muon2017_bin7_VG
785 prop_binCROS_electron2016M_bin6_TTTo2L2Nu
786 prop_binCROS_muon2018_bin14_TTTo2L2Nu
787 prop_binCROS_electron2016M_bin9_TTTo2L2Nu
788 prop_binCROS_electron2018_bin5_TTTo2L2Nu
789 prop_binCRF_electron2016M_bin19_WpWpJJ_QCD
790 prop_binCRF_muon2016M_bin8_WpWpJJ_QCD
791 prop_binSR_electron2017_bin20_ZZtoLep
792 prop_binCROS_muon2016M_bin13_TTTo2L2Nu
793 prop_binCROS_electron2017_bin10_WZ
794 prop_binCRF_electron2016M_bin19_TVX
795 prop_binCRF_electron2017_bin16_TVX
796 prop_binSR_electron2017_bin20_WrongSign
797 prop_binCRF_electron2016M_bin15_WpWpJJ_QCD
798 prop_binCROS_muon2017_bin5_VG
799 prop_binCROS_electron2017_bin8_WZ
800 prop_binCROS_muon2018_bin7_VG
801 prop_binSR_muon2017_bin15_TVX
802 prop_binCRF_muon2016M_bin13_WpWpJJ_QCD
803 prop_binCRF_muon2016M_bin18_Triboson
804 prop_binCROS_electron2016M_bin5_VG
805 prop_binCRF_electron2016M_bin16_Triboson
806 prop_binCRF_muon2016M_bin17_WpWpJJ_QCD
807 prop_binCRF_muon2016M_bin11_Triboson
808 prop_binCRF_electron2016M_bin15_Triboson
809 prop_binSR_electron2017_bin15_ZZtoLep
810 prop_binCROS_electron2017_bin14_VG
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cll1



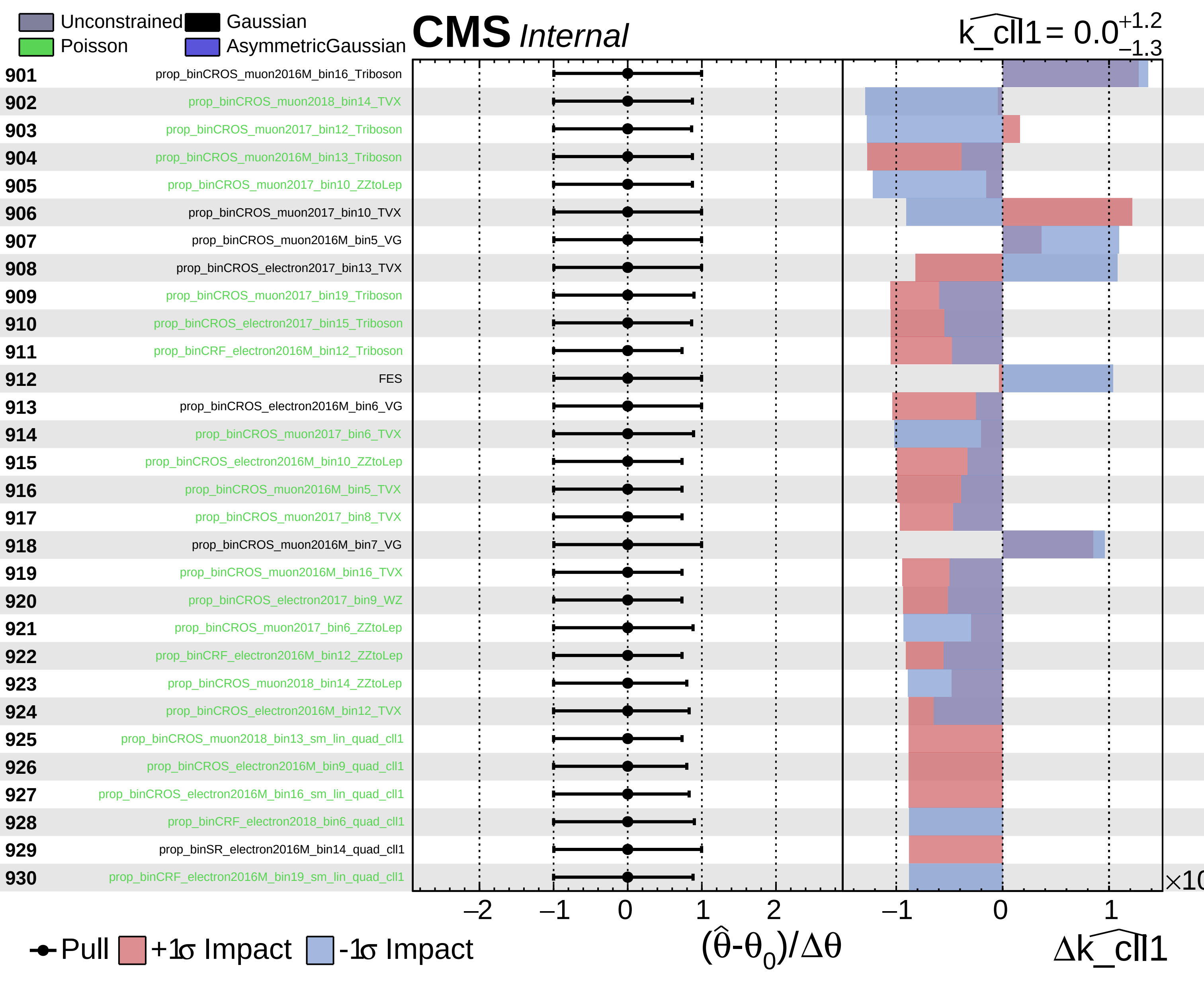
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
811 prop_binCROS_muon2018_bin15_TTTo2L2Nu
812 prop_binCROS_electron2018_bin18_TVX
813 prop_binCRF_muon2017_bin17_ZZtoLep
814 prop_binSR_muon2017_bin15_WpWpJJ_QCD
815 prop_binSR_electron2017_bin6_ZZtoLep
816 prop_binCRF_electron2017_bin18_ZZtoLep
817 prop_binCRF_muon2017_bin13_WpWpJJ_QCD
818 prop_binCRF_electron2018_bin18_WpWpJJ_QCD
819 prop_binCRF_electron2017_bin13_ZZtoLep
820 prop_binCRF_muon2016M_bin11_WpWpJJ_QCD
821 prop_binSR_electron2016M_bin18_ZZtoLep
822 prop_binCRF_electron2016M_bin11_TVX
823 prop_binCRF_muon2016M_bin15_Triboson
824 prop_binSR_muon2017_bin15_Triboson
825 prop_binSR_muon2016M_bin19_ZZtoLep
826 prop_binCRF_muon2016M_bin8_Triboson
827 prop_binCRF_electron2016M_bin16_ZZtoLep
828 prop_binCRF_muon2016M_bin12_Triboson
829 prop_binSR_muon2016M_bin16_ZZtoLep
830 prop_binCRF_muon2017_bin9_Triboson
831 prop_binCROS_muon2016M_bin7_TTTo2L2Nu
832 prop_binCRF_electron2018_bin18_Triboson
833 prop_binCRF_electron2016M_bin11_WrongSign
834 prop_binCROS_electron2016M_bin11_TTTo2L2Nu
835 prop_binCRF_electron2018_bin6_sm
836 prop_binCRF_electron2016M_bin18_ZZtoLep
837 prop_binCRF_electron2018_bin14_Triboson
838 prop_binCRF_electron2016M_bin19_Triboson
839 prop_binCRF_muon2016M_bin17_Triboson
840 prop_binCRF_muon2016M_bin12_ZZtoLep
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cll1



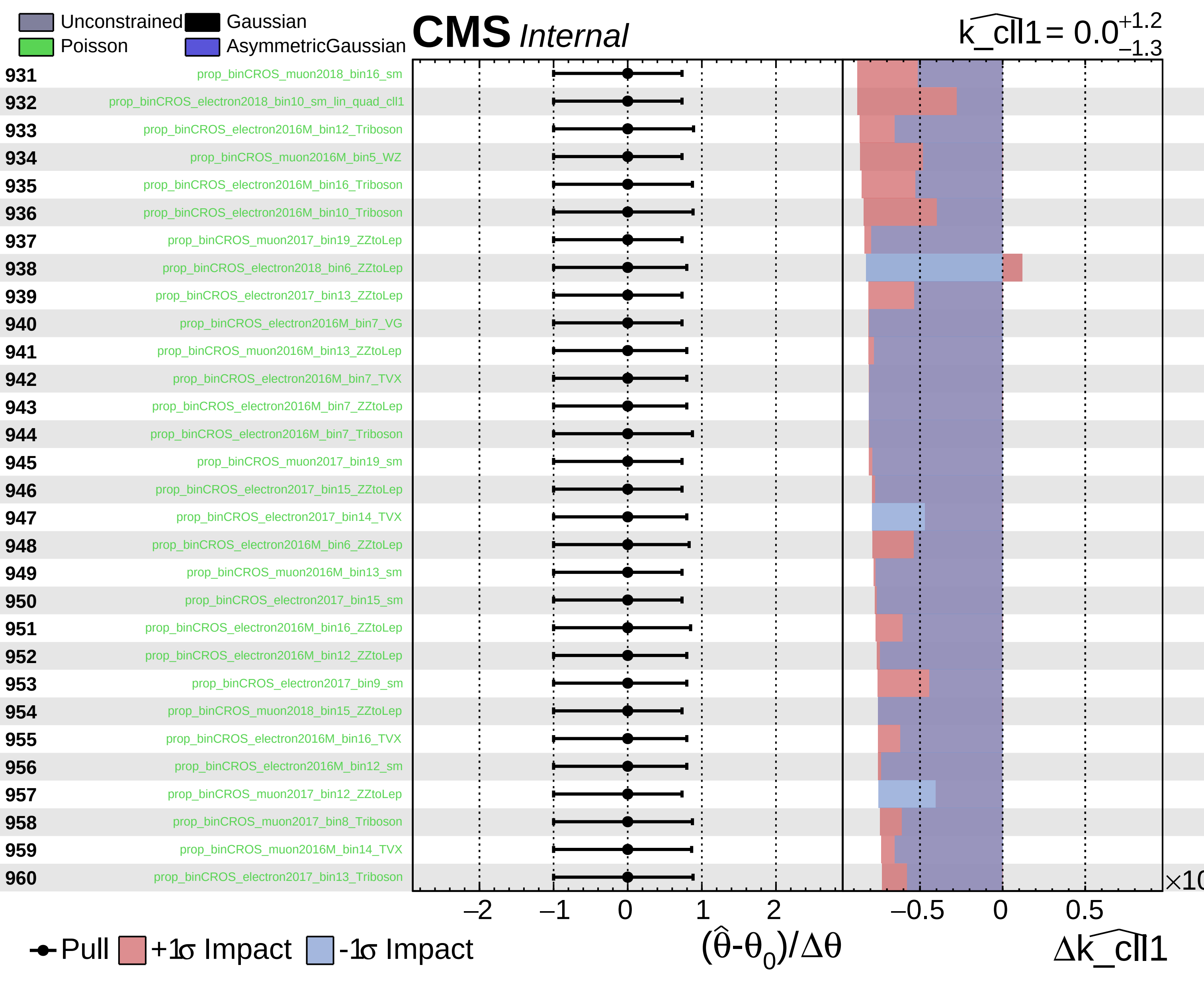
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
841 prop_binCRF_electron2018_bin6_ZZtoLep
842 prop_binCROS_muon2017_bin14_TVX
843 prop_binCRF_electron2017_bin13_Triboson
844 prop_binCRF_muon2016M_bin8_ZZtoLep
845 prop_binCROS_electron2017_bin13_Fake
846 prop_binCRF_electron2016M_bin19_ZZtoLep
847 prop_binCRF_electron2017_bin16_Triboson
848 prop_binCROS_muon2018_bin14_VG
849 prop_binCRF_electron2017_bin16_ZZtoLep
850 prop_binCRF_electron2016M_bin15_ZZtoLep
851 prop_binCRF_muon2016M_bin13_WrongSign
852 prop_binCRF_electron2018_bin6_TVX
853 prop_binCROS_electron2018_bin5_TVX
854 prop_binCRF_muon2016M_bin15_ZZtoLep
855 prop_binCROS_muon2017_bin10_VG
856 prop_binCROS_electron2017_bin13_TTTo2L2Nu
857 prop_binCRF_electron2016M_bin12_sm
858 prop_binCRF_muon2016M_bin11_ZZtoLep
859 prop_binCRF_electron2016M_bin12_TVX
860 prop_binCROS_electron2017_bin14_Triboson
861 prop_binCROS_muon2017_bin7_TVX
862 prop_binCRF_electron2018_bin6_WpWpJJ_QCD
863 prop_binCRF_electron2018_bin6_Triboson
864 prop_binCROS_muon2018_bin5_TVX
865 prop_binCROS_muon2018_bin11_TVX
866 prop_binCROS_muon2018_bin14_Triboson
867 prop_binCROS_muon2018_bin15_Triboson
868 prop_binCROS_electron2016M_bin11_TVX
869 prop_binCROS_muon2018_bin5_Triboson
870 prop_binCROS_muon2017_bin6_VG
−2 −1 0 1 2
−0.02 0 0.02
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cll1



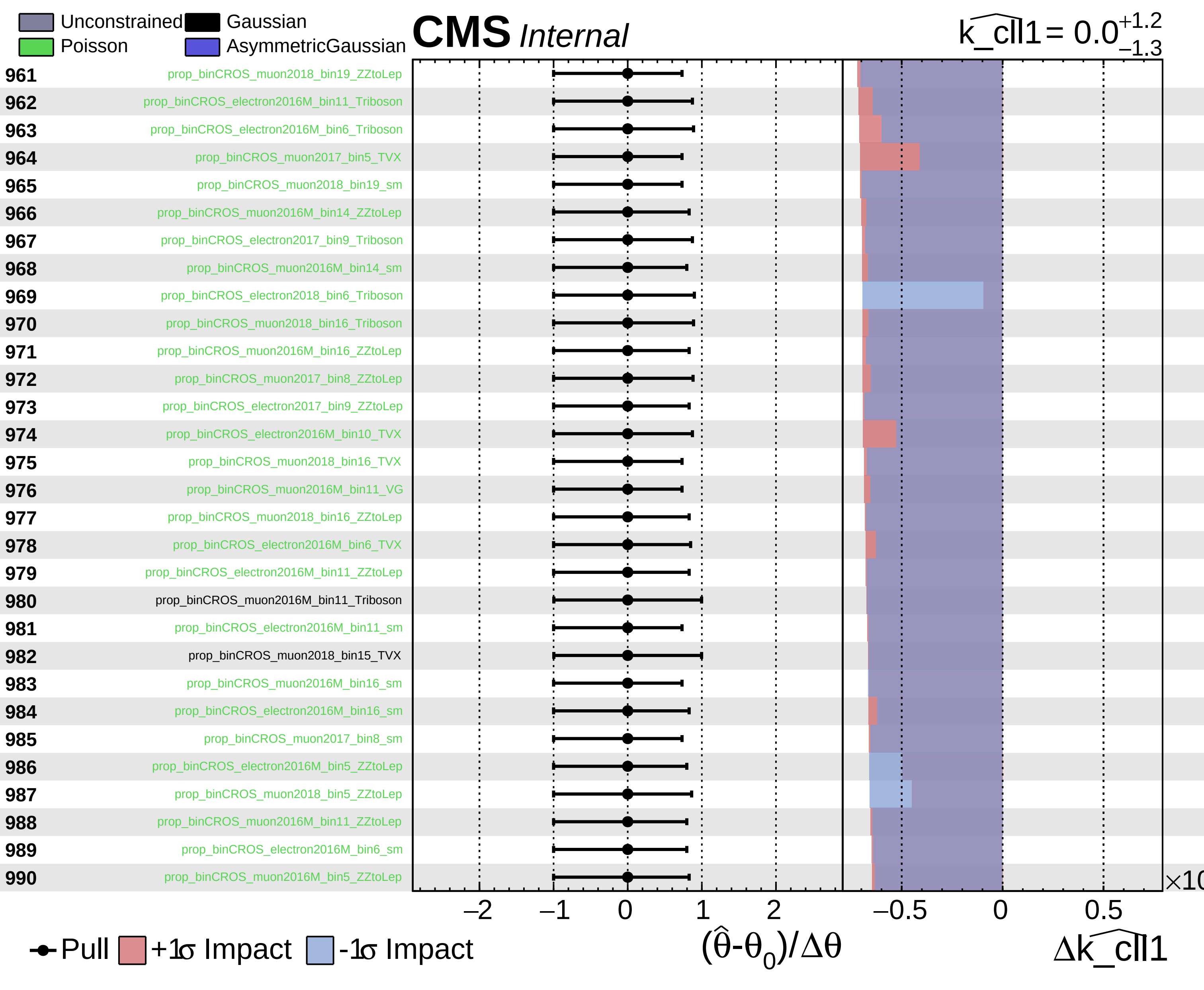
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = $0.0^{+1.2}_{-1.3}$
871 prop_binCROS_muon2017_bin14_Triboson
872 prop_binCROS_electron2018_bin5_ZZtoLep
873 prop_binCROS_muon2018_bin19_VG
874 prop_binCROS_muon2018_bin7_Triboson
875 prop_binCROS_electron2018_bin5_Triboson
876 prop_binCROS_muon2017_bin8_VG
877 prop_binCROS_electron2018_bin6_TVX
878 prop_binCROS_electron2016M_bin12_Fake
879 prop_binCRF_electron2016M_bin12_WpWpJJ_QCD
880 prop_binCROS_muon2017_bin10_Triboson
881 prop_binCROS_muon2017_bin12_TVX
882 prop_binCROS_muon2018_bin11_Triboson
883 prop_binCROS_muon2017_bin6_Triboson
884 prop_binCROS_muon2017_bin7_ZZtoLep
885 prop_binCROS_muon2018_bin7_TVX
886 prop_binCROS_muon2018_bin19_Triboson
887 prop_binCROS_muon2017_bin19_TVX
888 prop_binCROS_muon2018_bin13_Triboson
889 prop_binCROS_muon2018_bin19_TVX
890 prop_binCROS_muon2016M_bin13_TVX
891 prop_binCROS_muon2017_bin7_Triboson
892 prop_binCROS_electron2016M_bin5_TVX
893 prop_binCROS_electron2016M_bin10_VG
894 prop_binCROS_electron2017_bin15_TVX
895 prop_binCROS_electron2016M_bin5_Triboson
896 prop_binCROS_electron2018_bin8_Triboson
897 prop_binCROS_muon2018_bin7_ZZtoLep
898 prop_binCROS_electron2017_bin9_VG
899 prop_binCROS_electron2016M_bin6_WZ
900 prop_binCROS_muon2017_bin14_ZZtoLep
−2 −1 0 1 2
−2 0 2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cll1



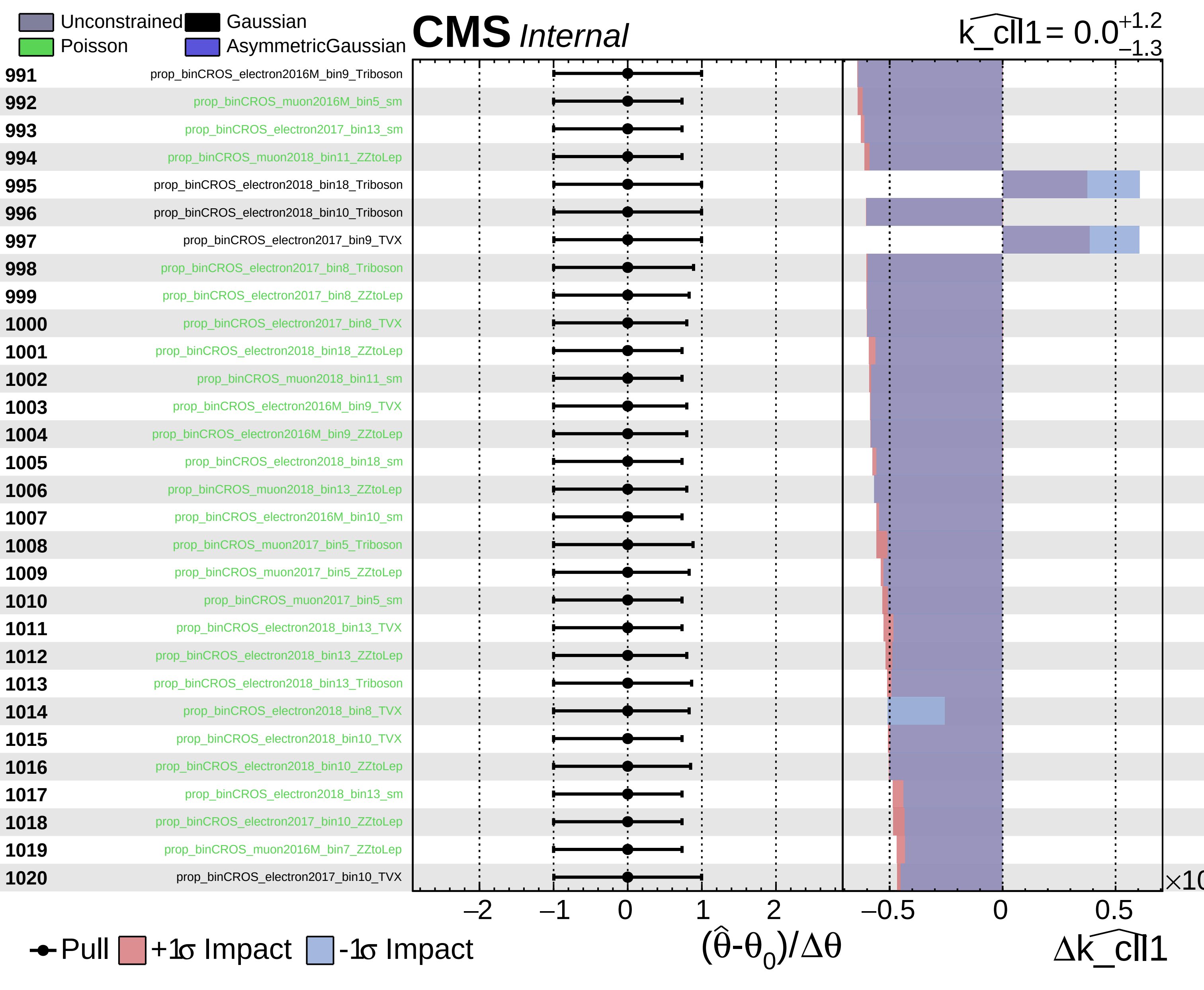
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cll1} = 0.0^{+1.2}_{-1.3}$
901 prop_binCROS_muon2016M_bin16_Triboson
902 prop_binCROS_muon2018_bin14_TVX
903 prop_binCROS_muon2017_bin12_Triboson
904 prop_binCROS_muon2016M_bin13_Triboson
905 prop_binCROS_muon2017_bin10_ZZtoLep
906 prop_binCROS_muon2017_bin10_TVX
907 prop_binCROS_muon2016M_bin5_VG
908 prop_binCROS_electron2017_bin13_TVX
909 prop_binCROS_muon2017_bin19_Triboson
910 prop_binCROS_electron2017_bin15_Triboson
911 prop_binCRF_electron2016M_bin12_Triboson
912 FES
913 prop_binCROS_electron2016M_bin6_VG
914 prop_binCROS_muon2017_bin6_TVX
915 prop_binCROS_electron2016M_bin10_ZZtoLep
916 prop_binCROS_muon2016M_bin5_TVX
917 prop_binCROS_muon2017_bin8_TVX
918 prop_binCROS_muon2016M_bin7_VG
919 prop_binCROS_muon2016M_bin16_TVX
920 prop_binCROS_electron2017_bin9_WZ
921 prop_binCROS_muon2017_bin6_ZZtoLep
922 prop_binCRF_electron2016M_bin12_ZZtoLep
923 prop_binCROS_muon2018_bin14_ZZtoLep
924 prop_binCROS_electron2016M_bin12_TVX
925 prop_binCROS_muon2018_bin13_sm_lin_quad_cll1
926 prop_binCROS_electron2016M_bin9_quad_cll1
927 prop_binCROS_electron2016M_bin16_sm_lin_quad_cll1
928 prop_binCRF_electron2018_bin6_quad_cll1
929 prop_binSR_electron2016M_bin14_quad_cll1
930 prop_binCRF_electron2016M_bin19_sm_lin_quad_cll1
−2
−1
0
1
2
−1
0
1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cll1}$



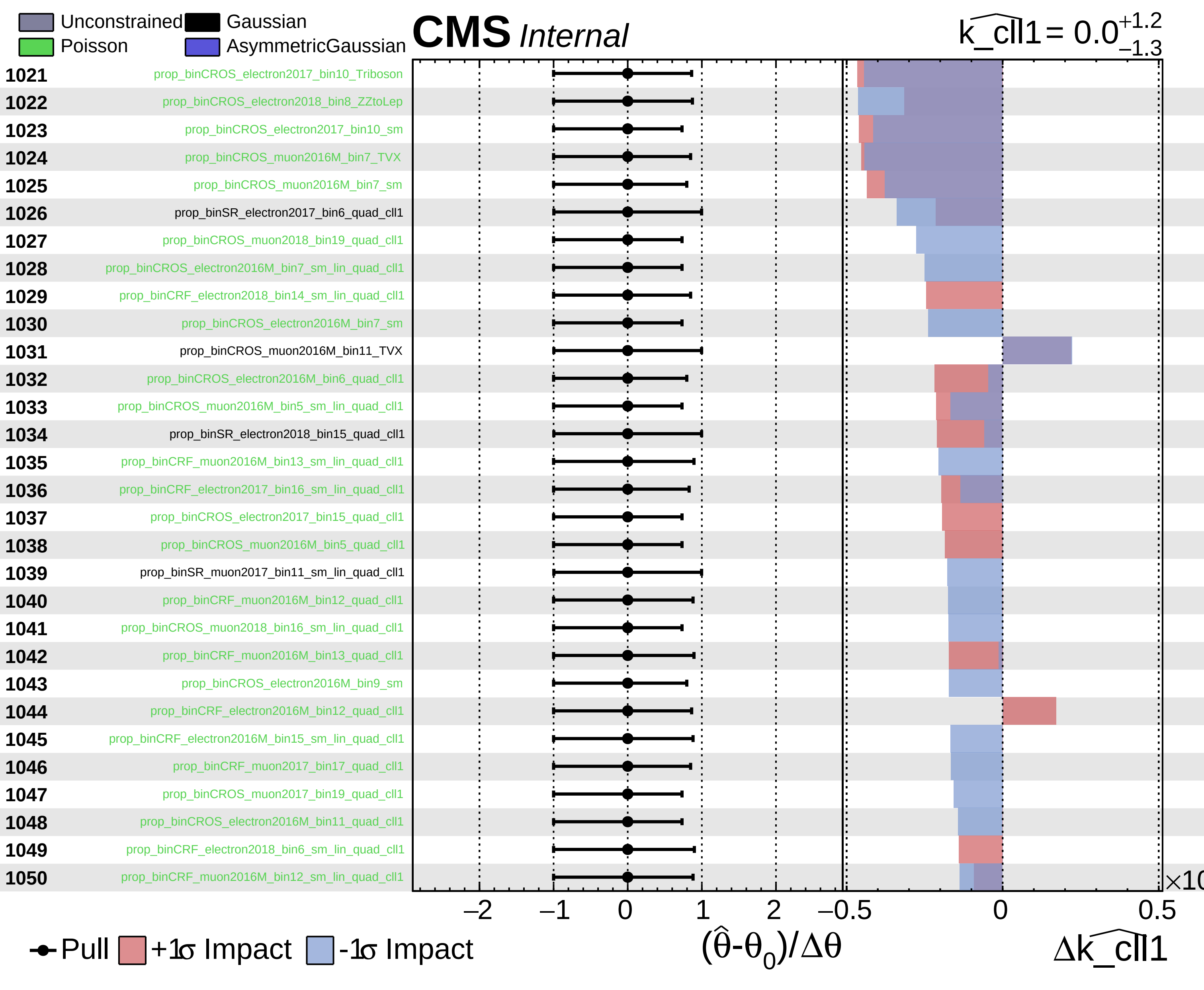
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
931 prop_binCROS_muon2018_bin16_sm
932 prop_binCROS_electron2018_bin10_sm_lin_quad_cll1
933 prop_binCROS_electron2016M_bin12_Triboson
934 prop_binCROS_muon2016M_bin5_WZ
935 prop_binCROS_electron2016M_bin16_Triboson
936 prop_binCROS_electron2016M_bin10_Triboson
937 prop_binCROS_muon2017_bin19_ZZtoLep
938 prop_binCROS_electron2018_bin6_ZZtoLep
939 prop_binCROS_electron2017_bin13_ZZtoLep
940 prop_binCROS_electron2016M_bin7_VG
941 prop_binCROS_muon2016M_bin13_ZZtoLep
942 prop_binCROS_electron2016M_bin7_TVX
943 prop_binCROS_electron2016M_bin7_ZZtoLep
944 prop_binCROS_electron2016M_bin7_Triboson
945 prop_binCROS_muon2017_bin19_sm
946 prop_binCROS_electron2017_bin15_ZZtoLep
947 prop_binCROS_electron2017_bin14_TVX
948 prop_binCROS_electron2016M_bin6_ZZtoLep
949 prop_binCROS_muon2016M_bin13_sm
950 prop_binCROS_electron2017_bin15_sm
951 prop_binCROS_electron2016M_bin16_ZZtoLep
952 prop_binCROS_electron2016M_bin12_ZZtoLep
953 prop_binCROS_electron2017_bin9_sm
954 prop_binCROS_muon2018_bin15_ZZtoLep
955 prop_binCROS_electron2016M_bin16_TVX
956 prop_binCROS_electron2016M_bin12_sm
957 prop_binCROS_muon2017_bin12_ZZtoLep
958 prop_binCROS_muon2017_bin8_Triboson
959 prop_binCROS_muon2016M_bin14_TVX
960 prop_binCROS_electron2017_bin13_Triboson
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cll1



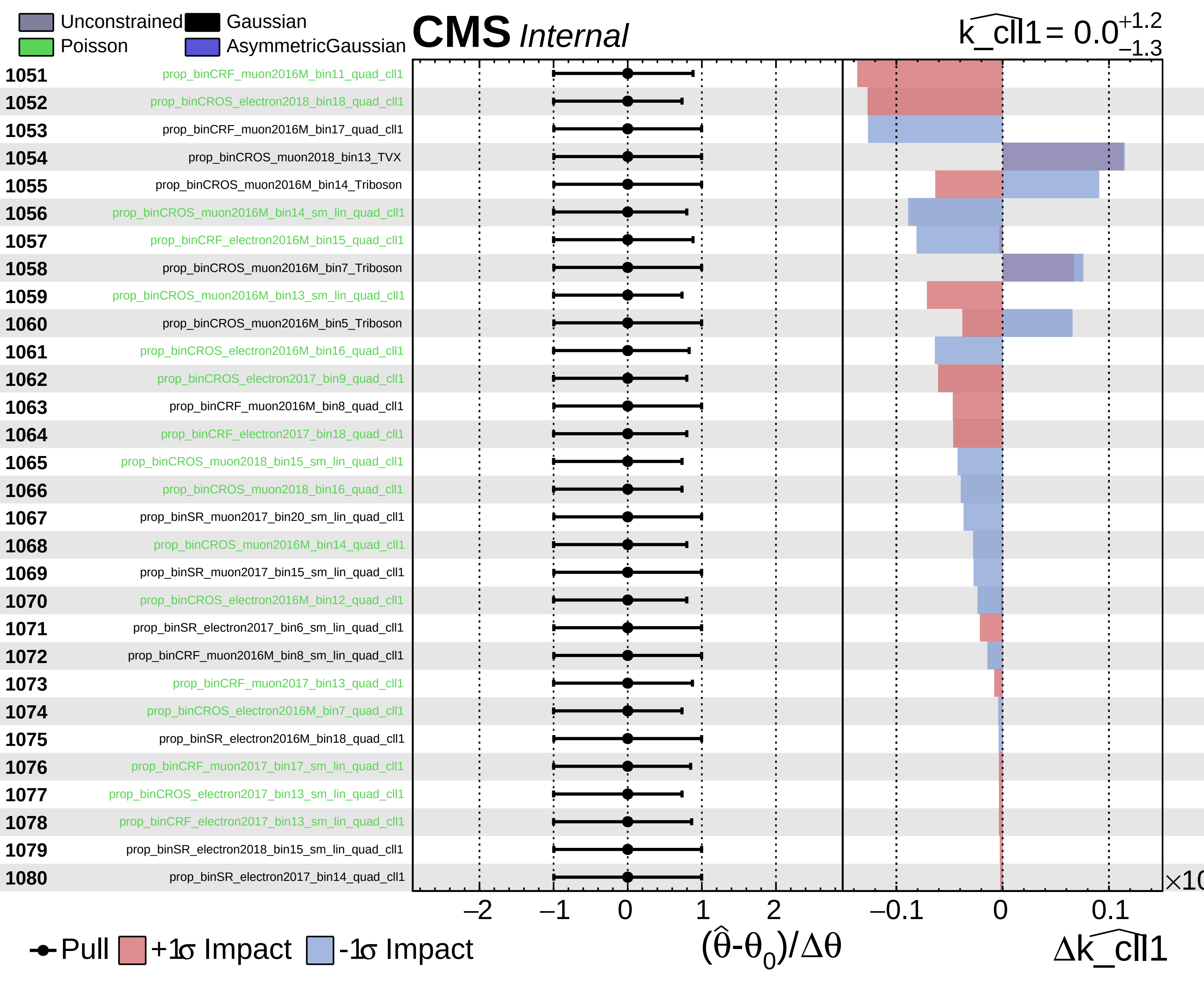
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
961 prop_binCROS_muon2018_bin19_ZZtoLep
962 prop_binCROS_electron2016M_bin11_Triboson
963 prop_binCROS_electron2016M_bin6_Triboson
964 prop_binCROS_muon2017_bin5_TVX
965 prop_binCROS_muon2018_bin19_sm
966 prop_binCROS_muon2016M_bin14_ZZtoLep
967 prop_binCROS_electron2017_bin9_Triboson
968 prop_binCROS_muon2016M_bin14_sm
969 prop_binCROS_electron2018_bin6_Triboson
970 prop_binCROS_muon2018_bin16_Triboson
971 prop_binCROS_muon2016M_bin16_ZZtoLep
972 prop_binCROS_muon2017_bin8_ZZtoLep
973 prop_binCROS_electron2017_bin9_ZZtoLep
974 prop_binCROS_electron2016M_bin10_TVX
975 prop_binCROS_muon2018_bin16_TVX
976 prop_binCROS_muon2016M_bin11_VG
977 prop_binCROS_muon2018_bin16_ZZtoLep
978 prop_binCROS_electron2016M_bin6_TVX
979 prop_binCROS_electron2016M_bin11_ZZtoLep
980 prop_binCROS_muon2016M_bin11_Triboson
981 prop_binCROS_electron2016M_bin11_sm
982 prop_binCROS_muon2018_bin15_TVX
983 prop_binCROS_muon2016M_bin16_sm
984 prop_binCROS_electron2016M_bin16_sm
985 prop_binCROS_muon2017_bin8_sm
986 prop_binCROS_electron2016M_bin5_ZZtoLep
987 prop_binCROS_muon2018_bin5_ZZtoLep
988 prop_binCROS_muon2016M_bin11_ZZtoLep
989 prop_binCROS_electron2016M_bin6_sm
990 prop_binCROS_muon2016M_bin5_ZZtoLep
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cll1



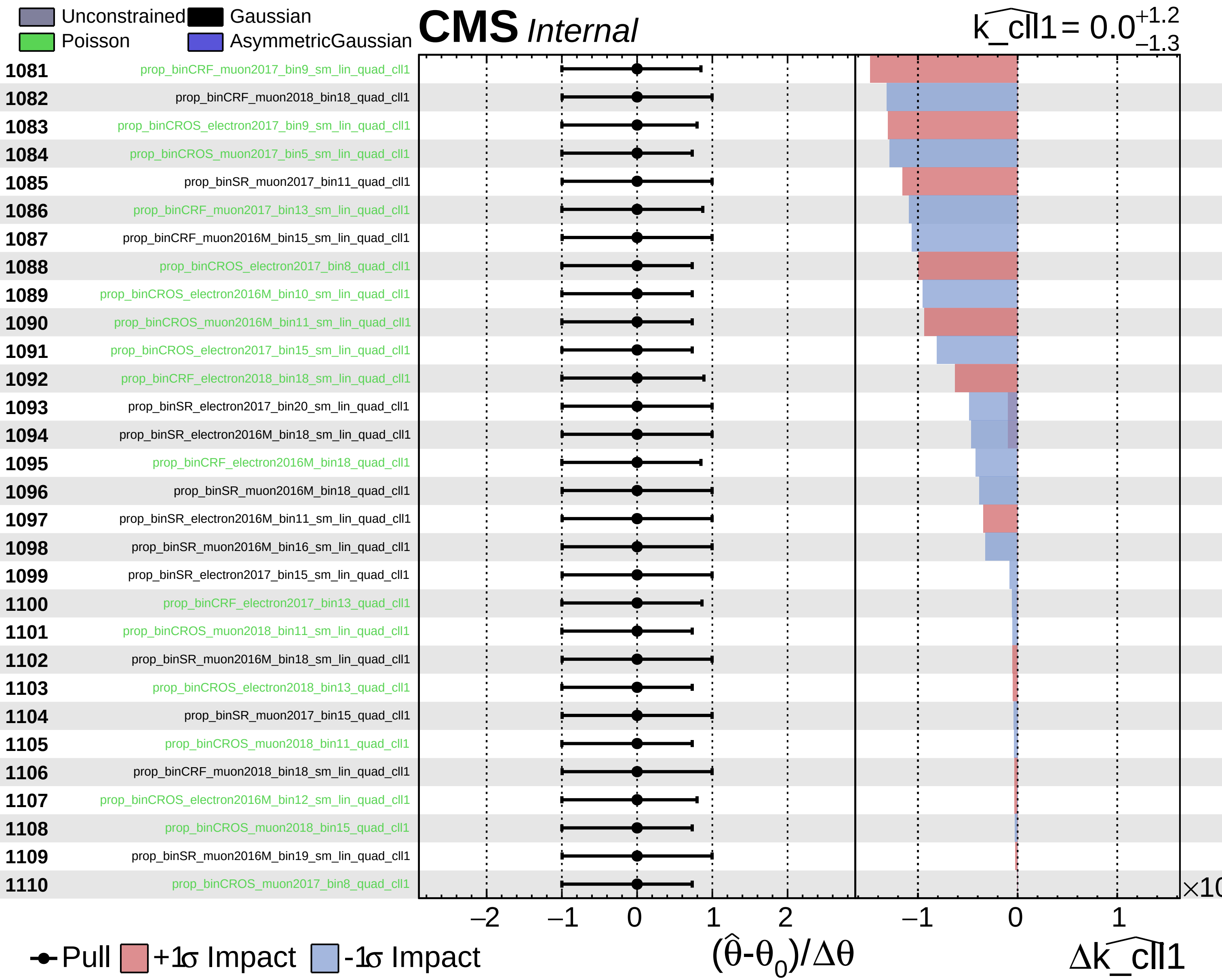
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
991 prop_binCROS_electron2016M_bin9_Triboson
992 prop_binCROS_muon2016M_bin5_sm
993 prop_binCROS_electron2017_bin13_sm
994 prop_binCROS_muon2018_bin11_ZZtoLep
995 prop_binCROS_electron2018_bin18_Triboson
996 prop_binCROS_electron2018_bin10_Triboson
997 prop_binCROS_electron2017_bin9_TVX
998 prop_binCROS_electron2017_bin8_Triboson
999 prop_binCROS_electron2017_bin8_ZZtoLep
1000 prop_binCROS_electron2017_bin8_TVX
1001 prop_binCROS_electron2018_bin18_ZZtoLep
1002 prop_binCROS_muon2018_bin11_sm
1003 prop_binCROS_electron2016M_bin9_TVX
1004 prop_binCROS_electron2016M_bin9_ZZtoLep
1005 prop_binCROS_electron2018_bin18_sm
1006 prop_binCROS_muon2018_bin13_ZZtoLep
1007 prop_binCROS_electron2016M_bin10_sm
1008 prop_binCROS_muon2017_bin5_Triboson
1009 prop_binCROS_muon2017_bin5_ZZtoLep
1010 prop_binCROS_muon2017_bin5_sm
1011 prop_binCROS_electron2018_bin13_TVX
1012 prop_binCROS_electron2018_bin13_ZZtoLep
1013 prop_binCROS_electron2018_bin13_Triboson
1014 prop_binCROS_electron2018_bin8_TVX
1015 prop_binCROS_electron2018_bin10_TVX
1016 prop_binCROS_electron2018_bin10_ZZtoLep
1017 prop_binCROS_electron2018_bin13_sm
1018 prop_binCROS_electron2017_bin10_ZZtoLep
1019 prop_binCROS_muon2016M_bin7_ZZtoLep
1020 prop_binCROS_electron2017_bin10_TVX
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cll1

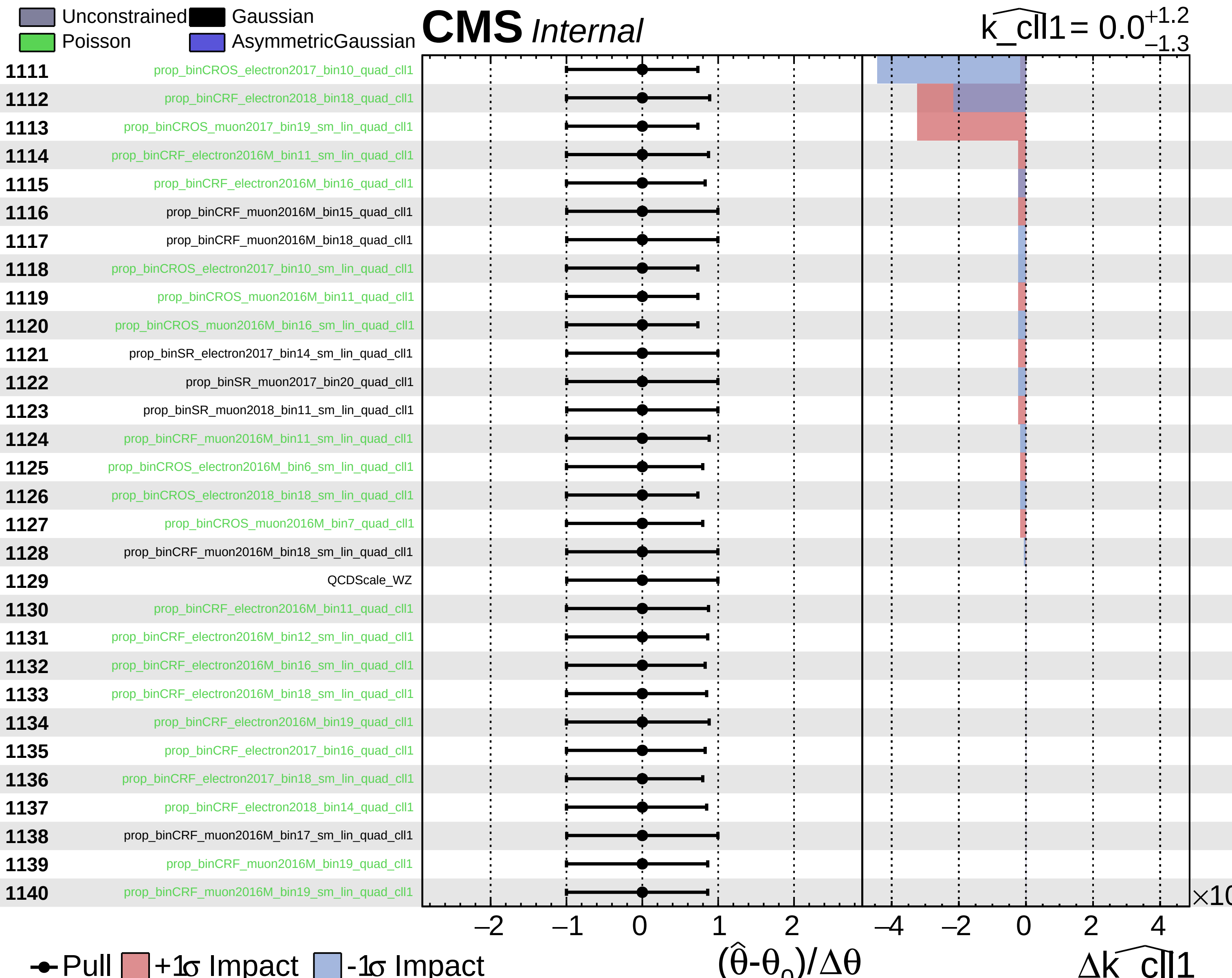


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
1021 prop_binCROS_electron2017_bin10_Triboson
1022 prop_binCROS_electron2018_bin8_ZZtoLep
1023 prop_binCROS_electron2017_bin10_sm
1024 prop_binCROS_muon2016M_bin7_TVX
1025 prop_binCROS_muon2016M_bin7_sm
1026 prop_binSR_electron2017_bin6_quad_cll1
1027 prop_binCROS_muon2018_bin19_quad_cll1
1028 prop_binCROS_electron2016M_bin7_sm_lin_quad_cll1
1029 prop_binCRF_electron2018_bin14_sm_lin_quad_cll1
1030 prop_binCROS_electron2016M_bin7_sm
1031 prop_binCROS_muon2016M_bin11_TVX
1032 prop_binCROS_electron2016M_bin6_quad_cll1
1033 prop_binCROS_muon2016M_bin5_sm_lin_quad_cll1
1034 prop_binSR_electron2018_bin15_quad_cll1
1035 prop_binCRF_muon2016M_bin13_sm_lin_quad_cll1
1036 prop_binCRF_electron2017_bin16_sm_lin_quad_cll1
1037 prop_binCROS_electron2017_bin15_quad_cll1
1038 prop_binCROS_muon2016M_bin5_quad_cll1
1039 prop_binSR_muon2017_bin11_sm_lin_quad_cll1
1040 prop_binCRF_muon2016M_bin12_quad_cll1
1041 prop_binCROS_muon2018_bin16_sm_lin_quad_cll1
1042 prop_binCRF_muon2016M_bin13_quad_cll1
1043 prop_binCROS_electron2016M_bin9_sm
1044 prop_binCRF_electron2016M_bin12_quad_cll1
1045 prop_binCRF_electron2016M_bin15_sm_lin_quad_cll1
1046 prop_binCRF_muon2017_bin17_quad_cll1
1047 prop_binCROS_muon2017_bin19_quad_cll1
1048 prop_binCROS_electron2016M_bin11_quad_cll1
1049 prop_binCRF_electron2018_bin6_sm_lin_quad_cll1
1050 prop_binCRF_muon2016M_bin12_sm_lin_quad_cll1
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cll1

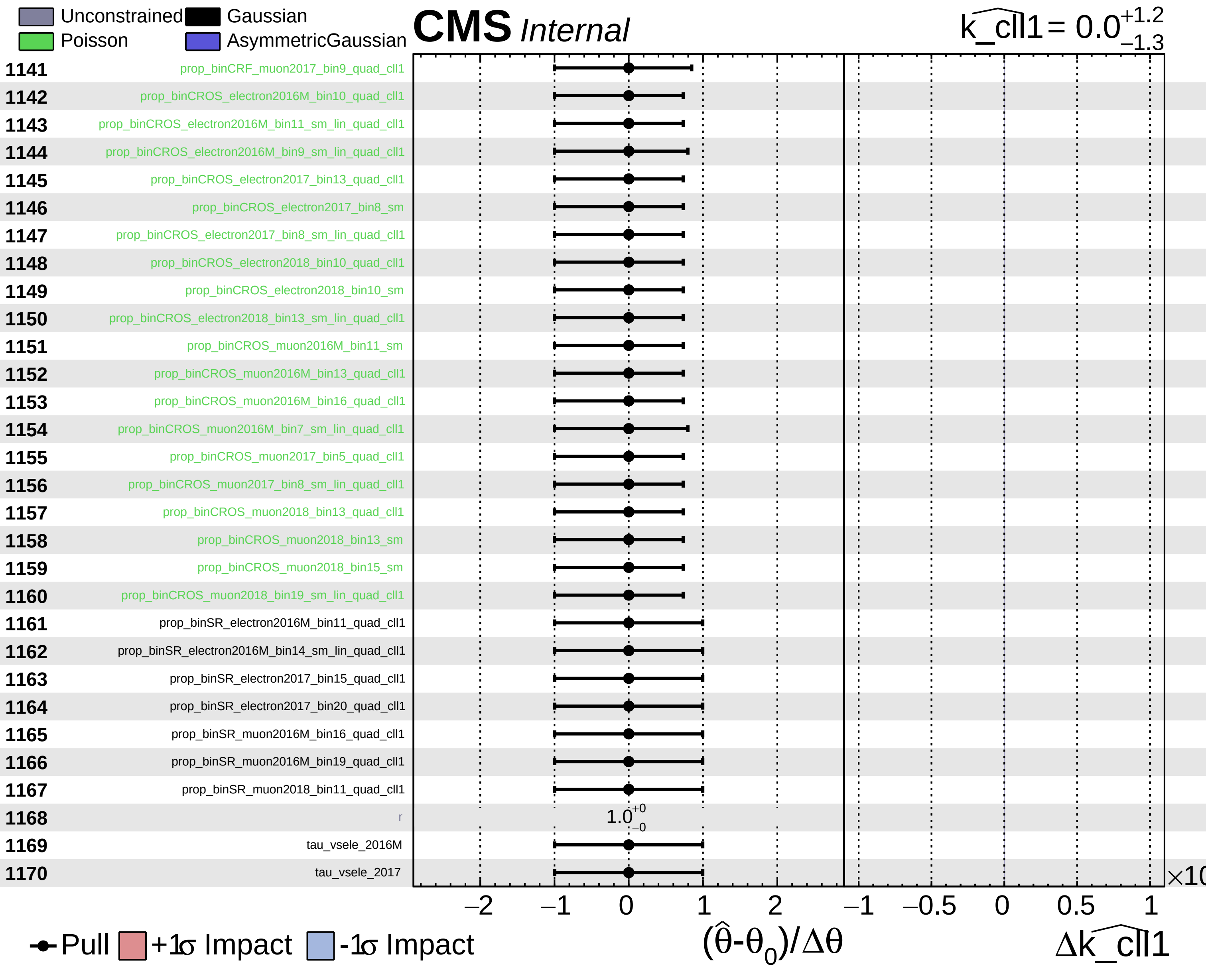


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
1051 prop_binCRF_muon2016M_bin11_quad_cll1
1052 prop_binCROS_electron2018_bin18_quad_cll1
1053 prop_binCRF_muon2016M_bin17_quad_cll1
1054 prop_binCROS_muon2018_bin13_TVX
1055 prop_binCROS_muon2016M_bin14_Triboson
1056 prop_binCROS_muon2016M_bin14_sm_lin_quad_cll1
1057 prop_binCRF_electron2016M_bin15_quad_cll1
1058 prop_binCROS_muon2016M_bin7_Triboson
1059 prop_binCROS_muon2016M_bin13_sm_lin_quad_cll1
1060 prop_binCROS_muon2016M_bin5_Triboson
1061 prop_binCROS_electron2016M_bin16_quad_cll1
1062 prop_binCROS_electron2017_bin9_quad_cll1
1063 prop_binCRF_muon2016M_bin8_quad_cll1
1064 prop_binCRF_electron2017_bin18_quad_cll1
1065 prop_binCROS_muon2018_bin15_sm_lin_quad_cll1
1066 prop_binCROS_muon2018_bin16_quad_cll1
1067 prop_binSR_muon2017_bin20_sm_lin_quad_cll1
1068 prop_binCROS_muon2016M_bin14_quad_cll1
1069 prop_binSR_muon2017_bin15_sm_lin_quad_cll1
1070 prop_binCROS_electron2016M_bin12_quad_cll1
1071 prop_binSR_electron2017_bin6_sm_lin_quad_cll1
1072 prop_binCRF_muon2016M_bin8_sm_lin_quad_cll1
1073 prop_binCRF_muon2017_bin13_quad_cll1
1074 prop_binCROS_electron2016M_bin7_quad_cll1
1075 prop_binSR_electron2016M_bin18_quad_cll1
1076 prop_binCRF_muon2017_bin17_sm_lin_quad_cll1
1077 prop_binCROS_electron2017_bin13_sm_lin_quad_cll1
1078 prop_binCRF_electron2017_bin13_sm_lin_quad_cll1
1079 prop_binSR_electron2018_bin15_sm_lin_quad_cll1
1080 prop_binSR_electron2017_bin14_quad_cll1
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cll1

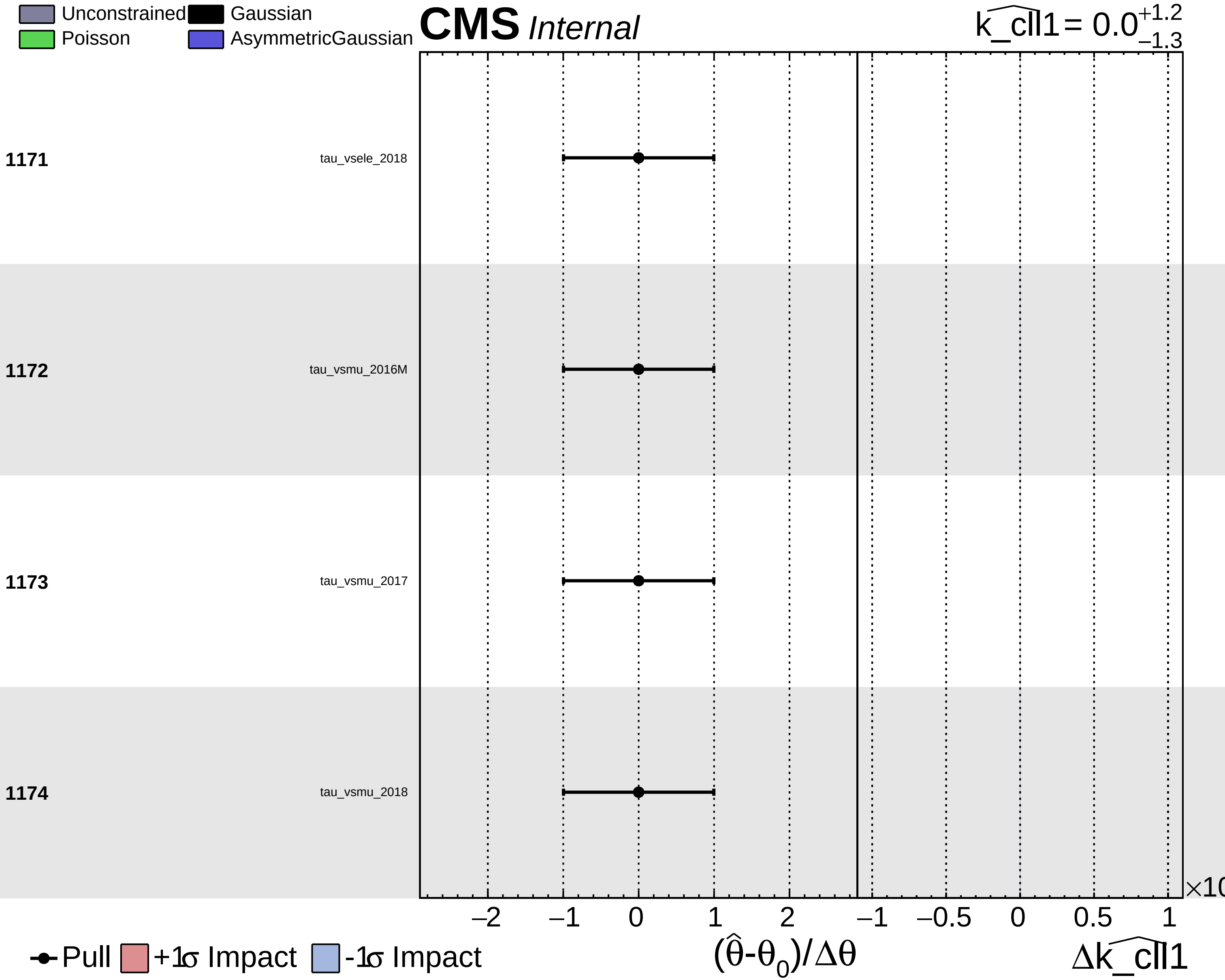

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cll1} = 0.0^{+1.2}_{-1.3}$
1081 prop_binCRF_muon2017_bin9_sm_lin_quad_cll1
1082 prop_binCRF_muon2018_bin18_quad_cll1
1083 prop_binCROS_electron2017_bin9_sm_lin_quad_cll1
1084 prop_binCROS_muon2017_bin5_sm_lin_quad_cll1
1085 prop_binSR_muon2017_bin11_quad_cll1
1086 prop_binCRF_muon2017_bin13_sm_lin_quad_cll1
1087 prop_binCRF_muon2016M_bin15_sm_lin_quad_cll1
1088 prop_binCROS_electron2017_bin8_quad_cll1
1089 prop_binCROS_electron2016M_bin10_sm_lin_quad_cll1
1090 prop_binCROS_muon2016M_bin11_sm_lin_quad_cll1
1091 prop_binCROS_electron2017_bin15_sm_lin_quad_cll1
1092 prop_binCRF_electron2018_bin18_sm_lin_quad_cll1
1093 prop_binSR_electron2017_bin20_sm_lin_quad_cll1
1094 prop_binSR_electron2016M_bin18_sm_lin_quad_cll1
1095 prop_binCRF_electron2016M_bin18_quad_cll1
1096 prop_binSR_muon2016M_bin18_quad_cll1
1097 prop_binSR_electron2016M_bin11_sm_lin_quad_cll1
1098 prop_binSR_muon2016M_bin16_sm_lin_quad_cll1
1099 prop_binSR_electron2017_bin15_sm_lin_quad_cll1
1100 prop_binCRF_electron2017_bin13_quad_cll1
1101 prop_binCROS_muon2018_bin11_sm_lin_quad_cll1
1102 prop_binSR_muon2016M_bin18_sm_lin_quad_cll1
1103 prop_binCROS_electron2018_bin13_quad_cll1
1104 prop_binSR_muon2017_bin15_quad_cll1
1105 prop_binCROS_muon2018_bin11_quad_cll1
1106 prop_binCRF_muon2018_bin18_sm_lin_quad_cll1
1107 prop_binCROS_electron2016M_bin12_sm_lin_quad_cll1
1108 prop_binCROS_muon2018_bin15_quad_cll1
1109 prop_binSR_muon2016M_bin19_sm_lin_quad_cll1
1110 prop_binCROS_muon2017_bin8_quad_cll1
−2
−1
0
1
2
−1
0
1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cll1}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cll1} = 0.0^{+1.2}_{-1.3}$
1111 prop_binCROS_electron2017_bin10_quad_cll1
1112 prop_binCRF_electron2018_bin18_quad_cll1
1113 prop_binCROS_muon2017_bin19_sm_lin_quad_cll1
1114 prop_binCRF_electron2016M_bin11_sm_lin_quad_cll1
1115 prop_binCRF_electron2016M_bin16_quad_cll1
1116 prop_binCRF_muon2016M_bin15_quad_cll1
1117 prop_binCRF_muon2016M_bin18_quad_cll1
1118 prop_binCROS_electron2017_bin10_sm_lin_quad_cll1
1119 prop_binCROS_muon2016M_bin11_quad_cll1
1120 prop_binCROS_muon2016M_bin16_sm_lin_quad_cll1
1121 prop_binSR_electron2017_bin14_sm_lin_quad_cll1
1122 prop_binSR_muon2017_bin20_quad_cll1
1123 prop_binSR_muon2018_bin11_sm_lin_quad_cll1
1124 prop_binCRF_muon2016M_bin11_sm_lin_quad_cll1
1125 prop_binCROS_electron2016M_bin6_sm_lin_quad_cll1
1126 prop_binCROS_electron2018_bin18_sm_lin_quad_cll1
1127 prop_binCROS_muon2016M_bin7_quad_cll1
1128 prop_binCRF_muon2016M_bin18_sm_lin_quad_cll1
1129 QCDScale_WZ
1130 prop_binCRF_electron2016M_bin11_quad_cll1
1131 prop_binCRF_electron2016M_bin12_sm_lin_quad_cll1
1132 prop_binCRF_electron2016M_bin16_sm_lin_quad_cll1
1133 prop_binCRF_electron2016M_bin18_sm_lin_quad_cll1
1134 prop_binCRF_electron2016M_bin19_quad_cll1
1135 prop_binCRF_electron2017_bin16_quad_cll1
1136 prop_binCRF_electron2017_bin18_sm_lin_quad_cll1
1137 prop_binCRF_electron2018_bin14_quad_cll1
1138 prop_binCRF_muon2016M_bin17_sm_lin_quad_cll1
1139 prop_binCRF_muon2016M_bin19_quad_cll1
1140 prop_binCRF_muon2016M_bin19_sm_lin_quad_cll1
−2 −1 0 1 2
−4 −2 0 2 4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cll1}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
1141 prop_binCRF_muon2017_bin9_quad_cll1
1142 prop_binCROS_electron2016M_bin10_quad_cll1
1143 prop_binCROS_electron2016M_bin11_sm_lin_quad_cll1
1144 prop_binCROS_electron2016M_bin9_sm_lin_quad_cll1
1145 prop_binCROS_electron2017_bin13_quad_cll1
1146 prop_binCROS_electron2017_bin8_sm
1147 prop_binCROS_electron2017_bin8_sm_lin_quad_cll1
1148 prop_binCROS_electron2018_bin10_quad_cll1
1149 prop_binCROS_electron2018_bin10_sm
1150 prop_binCROS_electron2018_bin13_sm_lin_quad_cll1
1151 prop_binCROS_muon2016M_bin11_sm
1152 prop_binCROS_muon2016M_bin13_quad_cll1
1153 prop_binCROS_muon2016M_bin16_quad_cll1
1154 prop_binCROS_muon2016M_bin7_sm_lin_quad_cll1
1155 prop_binCROS_muon2017_bin5_quad_cll1
1156 prop_binCROS_muon2017_bin8_sm_lin_quad_cll1
1157 prop_binCROS_muon2018_bin13_quad_cll1
1158 prop_binCROS_muon2018_bin13_sm
1159 prop_binCROS_muon2018_bin15_sm
1160 prop_binCROS_muon2018_bin19_sm_lin_quad_cll1
1161 prop_binSR_electron2016M_bin11_quad_cll1
1162 prop_binSR_electron2016M_bin14_sm_lin_quad_cll1
1163 prop_binSR_electron2017_bin15_quad_cll1
1164 prop_binSR_electron2017_bin20_quad_cll1
1165 prop_binSR_muon2016M_bin16_quad_cll1
1166 prop_binSR_muon2016M_bin19_quad_cll1
1167 prop_binSR_muon2018_bin11_quad_cll1
1168 r
1.0 +0 −0
1169 tau_vsele_2016M
1170 tau_vsele_2017
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cll1


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cll1 = 0.0 +1.2 −1.3
1171
tau_vsele_2018
1172
tau_vsmu_2016M
1173
tau_vsmu_2017
1174
tau_vsmu_2018
−2
−1
0
1
2
−1
−0.5
0
0.5
1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cll1